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Genomic Drivers of Organ Specific Metastasis and Brain Tropism in Breast Cancer: Metabolic Mechanisms and Nonlinear Time to Event Modeling with XGBoost and Accelerated Failure Time Approaches

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Abstract

This study synthesizes genomic, metabolic and time-to-event modeling perspectives to characterize organ-specific metastasis with emphasis on brain colonization by breast cancer and prognostic modeling in advanced lung cancer. Mechanistic analysis highlights adaptive metabolic programs in brain-tropic cells, including upregulation of de novo serine synthesis via PHGDH, enhanced glutamine and branched-chain amino acid oxidation, increased OXPHOS and antioxidant pathways, and lipogenic reprogramming driven by PTEN loss, RARRES2 suppression and SREBP1 activation; astrocyte and microglial crosstalk, gap junction mediated cyclic GMP–AMP transfer, EV mediated metabolic rewiring and BBB interactions are described as microenvironmental enablers of colonization. Translationally relevant vulnerabilities emerge in autophagy and stress response nodes (GRP94, KISS1), fatty acid handling (FABP7, CD36), and ferroptosis susceptibility (neratinib effects), suggesting combinatorial strategies that target metabolic flux, mitochondrial function and intercellular signalling. Methodological work compares Cox proportional hazards, Random Survival Forests and XGBoost with a Cox loss on stage IV cohorts, showing ensemble methods tuned by Bayesian optimization achieve higher concordance indices and lower Integrated Brier Scores than Cox and single trees, while emphasizing the need for rigorous calibration, explainability and external validation. The integrated framework proposed maps experimentally derived pathway markers into candidate covariates for survival models, recommends systematic hyperparameter optimization and resampling validation, and argues for multi center, molecularly annotated datasets to bridge mechanistic biology with prognostic tools able to capture nonlinear interactions between metastatic site, molecular programs and treatment variables.

1 Background and motivation

1.1 Clinical burden of organ-specific metastasis

Organ-specific metastasis imposes a substantial clinical burden because tumour cells must not only disseminate but also adapt to distinct metabolic and stromal milieus in distant organs. Metastatic breast cancer cells colonizing the brain face pronounced metabolic constraints, including low availability of glucose and amino acids, yet successfully establish brain metastases that remain a major cause of morbidity and mortality in advanced disease. Evidence indicates that brain metastatic breast cancer cells acquire metabolic signatures tailored for survival in the brain, including altered gluconeogenesis and oxidation of glutamine and branched-chain amino acids as alternative energy sources in the low-glucose environment. These cells display enhanced capacity to use glutamine both as an energy source and as a precursor for purines, pyrimidines and non-essential amino acids, suggesting that metabolic flexibility is closely tied to the ability to thrive in this organ. Autophagy-mediated mechanisms, such as KISS1-mediated survival autophagy and GRP94-driven unfolded protein response in low-glucose conditions, further support survival of breast cancer brain metastases and have been proposed as potential diagnostic markers and therapeutic targets in this setting [MH21; Ngo+20]. The brain microenvironment is characterized by complex metabolic coupling between astrocytes and neurons, which normally orchestrates glutamate, glutamine and lactate fluxes. Disruption of these circuits by metastatic breast cancer cells exacerbates clinical burden by enhancing tumour growth within the brain parenchyma. Extracellular vesicles (EVs) containing miR-199b derived from brain-tropic breast cancer cells downregulate key transporters, including EAAT2 in astrocytes and SNAT2 and MCT2 in excitatory neurons, thereby suppressing glutamate, glutamine and lactate consumption by brain cells. As a consequence, extracellular glutamine and lactate accumulate, and this retained pool of metabolites supports enhanced growth of breast cancer cells on organotypic brain slices. The growth-promoting effect of EVs with high miR-199b levels is lost when cancer cells are rendered unable to metabolize glutamine or lactate by silencing GLS or MCT1, underscoring that reprogramming of brain metabolic fluxes directly facilitates metastatic tumour expansion and likely contributes to neurological deterioration in affected patients [Rua+24]. Beyond the brain, metastatic organotropism is framed as a bidirectional metabolic adaptation process between cancer cells and the secondary organ. Cancer cells that metastasize to bone, liver, lung and brain must adjust to differences in energy and nutrient sources, organ-specific metabolites, pH, hypoxia, and cell–cell metabolic interactions. Primary tumours can remodel distant metabolic microenvironments during formation of premetastatic niches, while established macrometastases continue to reshape the secondary organ. This ongoing co-adaptation implies that organ-specific metastases are not static lesions but dynamic entities that modify systemic metabolism, contributing to weight loss, organ dysfunction, and treatment intolerance across the course of disease [WL21]. The clinical impact of organ-specific metastatic patterns is also evident in advanced non-small-cell lung cancer, where particular metastatic sites consistently emerge as important prognostic markers. In a stage IV cohort analyzed with both traditional and machine-learning survival models, pleural and liver metastases were repeatedly identified as critical variables influencing survival across Cox proportional hazards, Random Survival Forests and XGBoost models. Even though limited event numbers constrained statistical significance for some metastatic locations, convergence of feature importance across methods indicates that metastatic burden in specific organs is a key determinant of poor outcomes. Ensemble-based machine-learning models, particularly Random Survival Forests and XGBoost, achieved higher concordance indices and lower integrated Brier scores than Cox and single-tree models, highlighting their utility for capturing complex prognostic effects of metastatic patterns in multidimensional clinical datasets. Nonetheless, the single-institution nature of the dataset and lack of external validation temper the immediate clinical generalizability of these findings and emphasize the need for larger, multi-center studies to refine risk stratification according to metastatic site [Als+26].

1.2 Conceptual framework of metastatic organotropism

Metastatic organotropism can be framed as the capacity of tumour cells to match their adaptive potential to the selective pressures imposed by distinct organ microenvironments. In brain metastasis, a central feature of this adaptation is metabolic reprogramming that permits survival in a nutrient limited and often hypoxic niche. Triple negative breast cancer cells that efficiently colonize the brain upregulate metabolic pathways such as de novo serine synthesis, positioning 3 phosphoglycerate dehydrogenase (PHGDH) as a critical determinant of brain metastatic competence. Genetic silencing or pharmacological inhibition of PHGDH attenuates brain metastasis and improves survival in mouse models, whereas enforced expression of catalytically active PHGDH in non brain tropic cells promotes brain colonization, underscoring that nutrient availability in the target organ can dictate pathway dependence during metastatic evolution [Ngo+20]. Conceptually, organotropism emerges when metabolic traits of the primary tumour intersect with the metabolic architecture of the secondary organ. Brain tissues rely predominantly on glucose metabolism, with astrocytes consuming glucose and neurons preferentially oxidizing lactate. Breast tumours that undergo Warburg type metabolic reprogramming may therefore become particularly suited to exploit this environment, as glycolytic flux and associated biosynthetic pathways can be redirected to support survival under low glucose and restricted amino acid availability. Brain metastatic breast cancer cells further adapt by altering gluconeogenesis and oxidizing glutamine and branched chain amino acids as alternative energy sources, and by enhancing glutamine utilization for biosynthesis of nucleotides and non essential amino acids. These traits exemplify how metabolic flexibility can be viewed as a core axis of organ specific metastatic fitness [MH21; Ngo+20]. Beyond serine and glutamine metabolism, autophagy emerges as an auxiliary survival program that integrates metabolic stress signals within the brain microenvironment. Overexpression of the glucose regulated chaperone GRP94 relieves metabolic stress in low glucose conditions by coordinating the unfolded protein response and pro survival autophagy in brain metastases, while KISS1 mediated survival autophagy has been proposed as a diagnostic marker and potential therapeutic axis in this setting. Autophagy related regulators such as BECN1, FIP200, p53 and YAP contribute context dependent effects on tumour initiation, progression and metastatic dissemination, highlighting that organotropism is shaped not only by baseline metabolic wiring but also by stress response pathways that can be differentially engaged in specific organs. Conceptually, targeting autophagic flux becomes part of a broader strategy to disrupt the metabolic compatibility between metastatic cells and their preferred organ niches [MH21]. Metabolic organotropism is not restricted to the brain. In colorectal cancer, paired cell lines derived from a primary colon tumour (SW480) and a lymph node metastasis (SW620) illustrate stage specific reliance on lipid metabolism under nutrient stress. SW620 cells display greater dependence on lipid storage, and zinc borate disrupts lipid homeostasis and activates autophagy, revealing vulnerabilities that differ between primary and metastatic lesions within the same genetic background. This example reinforces the view that metastatic organotropism can be understood as context specific exploitation of lipid, amino acid or glucose metabolism, coupled to autophagy, in a fashion that reflects both tumour stage and organ location [KÖG26]. Time to event modeling contributes a complementary conceptual layer by quantifying how organ specific metastatic patterns and associated covariates affect survival. In stage IV non small cell lung cancer, Random Survival Forests and XGBoost survival models outperform Cox proportional hazards in terms of concordance index and integrated Brier score, capturing complex nonlinear interactions among clinical, metastatic and treatment variables. Although these analyses focus on advanced lung cancer, they demonstrate that ensemble machine learning methods can model intricate multidimensional prognostic links that are likely relevant for organ specific metastases, including brain involvement. Hyperparameter optimized RSF and XGBoost achieve lower prediction error than fixed parameter Cox or single tree models, framing organotropism not only as a biological phenomenon but also as a multidimensional prognostic construct that benefits from flexible survival modeling [Als+26].

1.3 Rationale for integrating genomics, metabolism, and survival modeling

Evidence from metastatic models and clinical survival analyses together motivates an integrated approach that unites genomic, metabolic and time to event modeling perspectives. In colorectal cancer, isogenic SW480 and SW620 cell lines originating from a primary colon tumour and a lymph node metastasis, respectively, have been exploited to dissect stage dependent metabolic adaptations under nutrient stress. Prior work cited in this context shows that SW620 cells rely more heavily on lipid metabolism when nutrients are limited, and zinc borate (ZnB) treatment was therefore tested for its capacity to perturb this dependence. ZnB reduced lipid droplet containing cells, disrupted lipid storage and altered lipid homeostasis, with effects that were particularly pronounced in SW620 cells, especially under glutamine deprivation. Combination with the mTOR inhibitor AZD 8055, an autophagy inducer, further decreased lipid droplet content and colony formation, suggesting that co targeting lipid metabolism and autophagy unmasks vulnerabilities that differ between primary and metastatic like cells. This staged comparison underscores that metabolic features, rather than purely genomic alterations, can stratify metastatic propensity and therapeutic sensitivity within an isogenic background [KÖG26]. Conceptual advances in organ specific metastasis strengthen this rationale by explicitly framing metastatic organotropism as a metabolic adaptation process that couples tumour cell plasticity to organ defined constraints. Analyses of cancer metabolism emphasize that malignant cells exhibit remarkable flexibility, including adoption of the Warburg effect and activation of alternative pathways such as ketone body utilization, modified fatty acid desaturation, and urea cycle rewiring. These adaptations enable energy generation, macromolecule synthesis and gene regulation under diverse environmental stresses. Within this framework, metabolic adaptation mechanisms are proposed as an emerging model of how cancer cells interact with and exploit host microenvironments, thereby contributing to organ specific colonization and growth [WL21]. Integrating genomic information about pathway components with functional metabolic phenotyping becomes essential for understanding which tumours can successfully colonize particular organs. Brain metastasis provides a salient example where metabolic rewiring in the secondary site intersects with organ specific constraints. The brain parenchyma is described as a high energy demand environment with low energy reserves, dependent on oxidative metabolism of glucose or liver derived ketone bodies. In contrast, the subarachnoid space is markedly hypoxic and nutrient deficient, creating distinct metabolic niches even within the central nervous system. Proteomic studies show that breast cancer cells metastasized to the brain upregulate enzymes involved in glycolysis, the tricarboxylic acid cycle, oxidative phosphorylation, the pentose phosphate pathway and the glutathione system. These alterations support oxidative metabolism of glucose and enhanced antioxidant capacity, thereby promoting survival in the oxidative and nutrient constrained brain milieu. Comparable enrichment of oxidative phosphorylation pathways in brain metastatic melanoma, and increased oxidative phosphorylation activity in intracranial compared with subcutaneous melanoma xenografts, further corroborate that mitochondrial energy metabolism and associated biosynthetic routes are selectively engaged during brain colonization. Infusion of ^{13}C labeled glucose in patients undergoing surgery for gliomas and brain metastases reveals that both entities use glucose for mitochondrial oxidation and for production of lactate, alanine, glutamic acid, glutamine and glycine, indicating coordinated engagement of glycolysis, the tricarboxylic acid cycle and non essential amino acid synthesis. Nonetheless, less than 50% of the acetyl CoA pool derives from blood glucose, implying that additional substrates contribute to energy metabolism in these lesions. Beyond glucose, lipid metabolism, exemplified by fatty acid binding protein 7, and amino acid metabolism, including gluconeogenesis and branched chain amino acid oxidation in brain metastatic breast cancer cells, are highlighted as critical determinants of adaptation and survival in the brain microenvironment [WL21; MH21]. Parallel efforts in survival modeling of advanced non small cell lung cancer further justify combining biological and methodological innovations. Traditional Cox proportional hazards models retain interpretability and offer semi parametric estimation of covariate effects under proportional hazards assumptions. However, this framework presupposes linear covariate effects on the log hazard and time invariant hazard ratios. Comparative analyses in stage IV cohorts illustrate that ensemble machine learning methods designed for survival data, including Random Survival Forests and XGBoost configured with a Cox loss, can better accommodate nonlinearities and higher order interactions among demographic, clinical, metastatic and treatment variables. In a single institution dataset with limited events, Random Survival Forest achieved the highest concordance index and the lowest integrated Brier score, with XGBoost showing comparable performance and both outperforming Cox and single decision tree models in repeated cross validation. Hyperparameter optimization with Bayesian strategies such as Optuna contributed to stable and unbiased performance estimates, while model reliability assessment using integrated Brier scores confirmed lower prediction errors for the ensemble based approaches. Nonetheless, decision tree and Cox models still provided clinically interpretable insights, and the absence of external validation and potential unmeasured confounders were recognized as important limitations, motivating larger, multi centre studies and hybrid strategies that balance predictive accuracy with interpretability [Als+26]. Taken together, these observations argue that a comprehensive framework for studying organ specific metastasis, and brain metastasis in particular, should integrate genomic determinants of metabolic pathways, functional metabolic adaptation in distinct organ microenvironments, and flexible survival modeling capable of capturing complex, potentially nonlinear associations among clinical, molecular and metastatic variables [KÖG26; WL21; MH21; Als+26].

2 Biology of metastatic dissemination and organ colonization

2.1 General principles of the metastatic cascade

Metastatic dissemination proceeds through a sequence of selective bottlenecks that collectively require cancer cells to tolerate therapy, remodel metabolism and eventually adapt to distinct organ microenvironments. In breast cancer, autophagy is repeatedly implicated as a central adaptive program operating across these stages. Autophagic flux sustains cell survival under hypoxia and nutrient deprivation by recycling intracellular components into amino acids, nucleotides, fatty acids, sugars and ATP. This enables tumour cells to endure the metabolic stress found in poorly perfused tumour cores and, by extension, in distant niches such as the brain, where glucose and other nutrients are limited. At early stages, autophagy can constrain necrosis and inflammation, thereby exerting anti metastatic effects, whereas at advanced stages it supports progression, therapeutic resistance and colonization of distal organs. Regulatory nodes including mTORC1, AKT and mitochondrial complex I integrate metabolic and stress signals to modulate autophagy, linking core bioenergetic pathways to dissemination capacity. As tumour cells exit the primary lesion and enter the circulation, they encounter additional constraints imposed by organ specific barriers. Brain colonization is particularly demanding because metastatic cells must traverse the blood–brain barrier and then withstand a nutrient deprived milieu. Evidence indicates that brain metastatic breast cancer cells adapt by rewiring metabolism to exploit alternative energy substrates and biosynthetic routes under low glucose conditions. These cells display altered gluconeogenesis and enhanced oxidation of glutamine and branched chain amino acids, and they use glutamine as both an energy source and a precursor for purines, pyrimidines and non essential amino acids. Autophagy complements these changes by maintaining metabolic homeostasis when exogenous nutrients are scarce, thereby providing a survival advantage during both transit and early seeding in the brain parenchyma. Once disseminated cells arrest in the microvasculature of a target organ, successful extravasation and niche formation depend on dynamic interactions with resident stromal and immune cells. In the brain, carcinoma–astrocyte gap junctions promoted by cancer cell protocadherin 7 (PCDH7) facilitate communication that activates interferon α and tumour necrosis factor signalling, with downstream activation of STAT1 and NF κB in cancer cells. These signalling cascades support brain metastasis and exemplify how stromal crosstalk integrates into the metastatic cascade. Parallel autophagy mediated mechanisms, such as KISS1 driven survival autophagy and GRP94 coordinated unfolded protein response, further enhance the ability of metastatic breast cancer cells to cope with brain specific stresses, consolidating colonization once micrometastatic lesions are established [MH21]. Later phases of the cascade involve expansion into clinically relevant macrometastases and systemic consequences. Metastatic lesions in secondary organs can reshape local and systemic metabolism, contributing to organ dysfunction and influencing patient survival, as discussed in Section 1.3. In advanced non small cell lung cancer, where most patients present with metastatic disease, survival outcomes reflect the combined impact of metastatic pattern, treatment and host factors. Cox proportional hazards models remain widely used to quantify covariate effects on time to event, but their assumptions of proportional hazards and linear covariate effects may not hold in complex datasets. Comparative analyses show that Random Survival Forests and XGBoost configured with a Cox loss achieve higher concordance indices and lower integrated Brier scores than Cox models and single decision trees when predicting survival in stage IV cohorts, indicating that ensemble methods can better capture nonlinear and interaction effects inherent to metastatic disease. Hyperparameter optimization with Bayesian strategies such as Optuna and evaluation via integrated Brier scores reinforce that these ensemble approaches provide more reliable probabilistic survival predictions, although interpretability considerations sustain a role for traditional regression models [Als+26]. Collectively, these observations portray the metastatic cascade as a multistep process in which metabolic plasticity, autophagy, stromal signalling and complex prognostic relationships converge. Autophagy and metabolic rewiring equip tumour cells to survive sequential bottlenecks from primary tumour escape to organ specific colonization, while modern survival modeling frameworks help quantify the prognostic impact of these processes in advanced metastatic settings [MH21; Als+26].

2.2 Brain microenvironment and blood-brain barrier

As metastatic breast cancer cells approach the central nervous system, they encounter the structural and cellular complexity of the blood–brain barrier (BBB). This barrier consists of endothelial cells connected by tight junctions, a thick basement membrane supported by pericytes, and an outer layer of astrocyte endfeet, collectively maintaining a tightly regulated, high electrical resistance interface that preserves a homeostatic milieu for neuronal function. Tight junctions are composed of proteins including claudins and occludins, whose integrity underpins the selective permeability of the BBB. The requirement to traverse this multilayered barrier renders brain colonization a particularly restrictive step in the metastatic cascade, and animal studies show that only a small fraction of circulating tumour cells can complete the process of distant colonization [MH21]. Mechanistic work on breast cancer brain metastasis describes BBB passage as a multistage process that is distinct from other metastatic sites. Breast cancer cells invade surrounding tissue, enter the circulation, and then lodge in brain microvessels, where they adhere to endothelial cells and the subendothelial matrix without overtly damaging vascular walls. Opening of tight junctions between tumour cells is described as a first step in their migration, and extravasation is influenced by endothelial cell activity and intercellular interactions rather than gross vascular disruption. Stromal proteases such as urokinase plasminogen activator and matrix metalloproteinase 2 facilitate passage through the basement membrane, while subsequent inhibition of plasmin by neuroserpin and serpin B2 allows tumour cells to cross the BBB and initiate colonization [Far+23]. Once across the BBB, metastatic cells encounter a brain microenvironment dominated by astrocytes, which normally support BBB integrity, deliver nutrients to neurons, regulate inflammatory responses, and protect neurons from hypoxia. Astrocytes express glucose transporters that enable glucose influx into the brain and glycoproteins that pump xenobiotics out, and they can dynamically modulate BBB tightness in response to stresses such as hypoxia. Under metastatic conditions, tumour cells evade astrocyte-mediated elimination and appropriate astrocyte functions for their own benefit. For example, breast cancer cells can secrete interleukin 1β or CCL2, enhancing differentiation of neural progenitor cells into astrocytes and generating reactive astrocytes characterized by glial fibrillary acidic protein upregulation. These reactive astrocytes produce a wide array of neurotrophic and inflammatory mediators, including TGFα, ATP, endothelin 1, glutamate, interleukins, TNFα, macrophage inflammatory protein 2, CCL2, CXCL12, glial cell derived neurotrophic factor, nitric oxide and sphingosine 1 phosphate, which collectively support metastatic growth and progression in the brain [MH21]. Gap junction mediated communication represents a critical signalling axis linking astrocytes and metastatic cells. After crossing the BBB, breast cancer cells exploit astrocyte gap junctions to transfer cyclic guanosine monophosphate adenosine monophosphate, inducing production of interferon α and tumour necrosis factor α in astrocytes. These mediators activate STAT1 and NFκB pathways in metastatic cells, promoting tumour growth and chemoresistance [Far+23]. Concordant findings attribute these interactions to cancer cell expression of protocadherin 7, which facilitates carcinoma–astrocyte gap junction formation and triggers interferon α and TNF production with downstream STAT1 and NFκB activation in cancer cells, thereby supporting brain metastasis [MH21]. These observations emphasize that the BBB is not merely a physical barrier but a signalling platform in which astrocytic responses are co-opted to enhance metastatic fitness. Additional components of the brain microenvironment, particularly microglia, further modulate metastatic dynamics. Estrogen, exemplified by 17β-estradiol, affects multiple brain-resident cell types and increases BBB permeability by altering inducible nitric oxide synthase activity and decreasing tight junction proteins such as occludin and claudin 1, thereby facilitating BBB disruption. In triple negative breast cancer models, premenopausal levels of estradiol promote brain metastasis through activation of brain derived neurotrophic factor and its receptor TrkB in reactive astrocytes, while tamoxifen displays antitumour effects in vivo largely by reprogramming microglial polarization toward an M1 phenotype and enhancing their phagocytic function. Estradiol also upregulates glutamate transporter expression and glutamate production in neurons, promoting brain tumour progression, and increases programmed death ligand 1 expression on microglia via STAT3, which suppresses T cell proliferation. In parallel, estradiol induces SIRPα on microglia and CD47 on tumour cells via PI3K/AKT signalling, strengthening the "do not eat me" axis that allows cancer cells to evade microglial phagocytosis; tamoxifen or pathway inhibition attenuates these effects [Wu+20]. Collectively, these data delineate a BBB associated ecosystem in which endothelial tight junctions, astrocytic support functions, and hormone sensitive microglial immune checkpoints are reprogrammed to enable breast cancer brain metastasis [MH21; Far+23; Wu+20].

2.3 Metabolic adaptation in nutrient-limited niches

Nutrient deprivation defines many metastatic niches, and cancer cells deploy diverse metabolic strategies to sustain proliferation under such constraints. In the brain, interstitial and cerebrospinal fluids are compositionally simple and markedly depleted in proteins and most amino acids relative to plasma, with glutamine as a notable exception, creating a microenvironment in which serine and glycine are particularly scarce. Within this context, triple negative breast cancer cells that efficiently colonize the brain upregulate de novo serine synthesis. Proteome comparisons identify 3 phosphoglycerate dehydrogenase (PHGDH), the rate limiting enzyme of glucose derived serine synthesis, as the most significantly upregulated protein in brain tropic cells. Genetic silencing or pharmacological inhibition of PHGDH attenuates brain metastasis and prolongs survival in mice, whereas enforced expression of catalytically active PHGDH in non brain tropic cells promotes brain colonization. These findings demonstrate that in serine and glycine limited environments, de novo serine synthesis becomes conditionally essential and that PHGDH represents a metabolic liability selectively exposed in the brain niche. Aligned with these mechanistic data, tailored culture media that recapitulate brain interstitial amino acid concentrations show that PHGDH dependence is enforced by environmental serine and glycine scarcity. Identical cancer cells growing in nutrient replete versus brain like media exhibit distinct metabolic programs and drug sensitivities, underscoring that nutrient availability modulates the phenotypic penetrance of metabolic alterations. In this setting, inhibition of PHGDH selectively impairs proliferation under brain mimetic conditions, indicating that amino acid limitation can be exploited to unmask therapeutic windows that would be less evident in standard culture media [Ngo+20]. Other studies extend the concept of metabolic adaptation in nutrient limited niches beyond serine metabolism. Analyses of brain metastases highlight that breast cancer cells in the parenchyma enhance expression of enzymes involved in glycolysis, the tricarboxylic acid cycle, oxidative phosphorylation (OXPHOS), the pentose phosphate pathway and the glutathione system. This configuration permits simultaneous oxidative metabolism of glucose and robust management of oxidative stress, thereby supporting survival in an environment characterized by low energy reserves but high energy demand. Infusion of ^{13}C labeled glucose into patients undergoing resection of brain gliomas and metastases shows that these tumours oxidize glucose in mitochondria and channel it into lactate, alanine, glutamic acid, glutamine and glycine, indicating coordinated activation of glycolysis, the TCA cycle, OXPHOS and non essential amino acid synthesis. Nonetheless, less than 50% of the acetyl CoA pool derives from blood glucose, implying that additional substrates, such as lipids or amino acids, contribute significantly to energy metabolism in these lesions. Lipid and amino acid pathways furnish further adaptive axes. High expression of fatty acid binding protein 7 (FABP7) correlates with poor prognosis and high incidence of brain metastasis in breast cancer patients. Functional work shows that FABP7 supports adaptation of HER2 positive breast cancer cells to the brain microenvironment by sustaining a glycolytic phenotype and promoting lipid droplet storage, linking fatty acid handling to energy production and fuel buffering under fluctuating nutrient supply. Moreover, brain metastatic breast cancer cells can invoke gluconeogenesis and branched chain amino acid oxidation to maintain growth in the nutrient constrained brain milieu, and acetate metabolism is also implicated in promoting brain metastases, although detailed mechanisms remain to be fully defined [WL21]. Nutrient stress induced reprogramming is not unique to brain lesions. In non small cell lung cancer, when glucose or serum is depleted, cells activate AMP activated protein kinase and switch from a predominantly glycolytic, Warburg like metabolism to reliance on the TCA cycle and OXPHOS, using glutamine and fatty acids as alternative carbon sources. Under glutamine deprivation, mutant KRAS enhances asparagine and aspartate metabolism via an Akt and NRF2 dependent axis, with NRF2 upregulating activating transcription factor 4 and asparagine synthetase, thereby sustaining biosynthesis in the absence of glutamine. Fatty acid oxidation is similarly engaged through ACC2 phosphorylation, supported by sufficient lipid storage. These adaptations illustrate how cancer cells dynamically rewire carbon fluxes among glycolysis, OXPHOS, amino acid and lipid metabolism to maintain ATP production and macromolecule synthesis during nutrient stress [Pei+25]. Parallel observations in colorectal cancer derived SW620 cells show that metastatic like cells possess higher energy metabolism capability than their primary tumour counterparts, with stronger OXPHOS and glycolysis potential and elevated expression of glycolytic regulators HK1, HK2, GLUT1 and HIF1α when glucose is sufficient. Suppression of OXPHOS associates intracellular ATP levels with sustained HK1 and HK2 expression, indicating that increased glycolytic capacity buffers against mitochondrial inhibition. Together, these findings reveal that metastatic and advanced cancer cells converge on flexible metabolic strategies, enabling survival in diverse nutrient limited niches by reconfiguring glycolytic, mitochondrial, amino acid and lipid pathways [Che+18].

2.4 Host factors shaping organ-specific colonization

Host-related microenvironmental features substantially influence whether disseminated tumour cells can successfully colonize distant organs. One major axis involves the metabolic landscape of potential target tissues. Pan cancer analyses comparing metabolic gene expression between primary tumours and their corresponding normal tissues show that malignant cells largely retain the metabolic programme of their tissue of origin, implying that host tissue identity constrains and shapes cancer metabolic patterns. Moreover, experimental work demonstrates that the same oncogenic driver can imprint distinct metabolic phenotypes depending on tissue context: MYC driven liver tumours exhibit reduced glutamine anabolism, whereas MYC driven lung tumours show increased glutamine utilization, and KRAS activation combined with additional genetic alterations produces divergent branched chain amino acid metabolism in mouse non small cell lung cancer versus pancreatic cancer. These observations underscore that organ specific metabolic baselines act as host factors that modulate how oncogenic lesions translate into metastatic fitness. Environmental differences between in vivo and in vitro systems, and between animal and human hosts, further illustrate how host conditions tune colonization capacity. Ras driven non small cell lung cancers rely predominantly on glucose to supply the tricarboxylic acid cycle in vivo, but glutamine becomes the main anaplerotic source in culture, highlighting that nutrient composition in the host circulation dictates metabolic routing. Uric acid concentrations, which are an order of magnitude higher in human than in mouse plasma, directly inhibit uridine monophosphate synthase and reduce sensitivity to 5 fluorouracil in human tumours, whereas this host factor is largely absent in mouse models because of lower uric acid levels. Intra tumoural heterogeneity within a single organ also reflects host perfusion differences: multimodality imaging combined with intraoperative ^{13}C glucose infusions reveals that differently perfused regions of non small cell lung cancer display distinct metabolic patterns, and both gliomas and brain metastases in the same brain environment can be fuelled by oxidized glucose and acetate without glutamine oxidation. Collectively, these data demonstrate that systemic and local host metabolism can either restrict or support metastatic colonization depending on nutrient availability and metabolite concentrations. Host organ microenvironments not only impose constraints but can actively drive metastatic relocation according to metabolic compatibility. Hypoxic lung cancer cells can leave the comparatively oxygen rich lung and metastasize to more hypoxic environments that better match their metabolic wiring, while liver cancer cells with vigorous mitochondrial metabolism may escape the hypoxic liver microenvironment and colonize the more oxygenated lung. This pattern indicates that gradients in oxygen tension and mitochondrial substrate availability across organs serve as host level selectors of metastatic destinations. The metabolic characteristics of specific tissues can thus be viewed as a form of "soil" that attracts "seeds" whose metabolic phenotype is better aligned with local conditions, reinforcing the need to map organ specific metabolic architectures when interpreting metastatic organotropism. Secondary sites themselves contain specialized stromal populations that shape colonization through metabolic crosstalk. In the omentum, adipocytes and cancer associated fibroblasts modulate ovarian cancer metastasis by supplying lipids and reprogramming glycogen metabolism. Coculture models show that activation of p38α MAPK in omentum derived fibroblasts leads to phosphorylation and activation of glucose phosphate mutase 1 in ovarian cancer cells under normoxia, increasing glycogenolysis and feeding glycolysis to promote proliferation, invasion and metastatic spread. In vivo, depletion of p38α in fibroblasts or pharmacologic inhibition of glycogenolysis reduces omental metastasis, demonstrating that host stromal signalling can establish a metabolically supportive niche that favours colonization. Lymph nodes provide another example of host determinants of organ specific colonization, with their microenvironment promoting lipid based metabolic reprogramming in metastatic cells. In melanoma and breast cancer metastases to lymph nodes, accumulated bile acids activate YAP via the vitamin D receptor, driving fatty acid oxidation and facilitating adaptation to the lymph node milieu. Long chain noncoding RNA LNMICC recruits NPM1 to the FABP5 promoter, reshaping fatty acid metabolism and promoting cervical cancer cell lymph node metastasis through epithelial–mesenchymal transition and VEGF C mediated lymphangiogenesis. Pharmacologic inhibition of fatty acid oxidation with etomoxir effectively suppresses lymph node metastasis, highlighting that host derived bile acids, stromal lipid handling and vascular remodelling collectively create a permissive metabolic environment for colonization that can be therapeutically targeted. Host factors also operate at the level of systemic circulation. Circulating tumour cells must match the metabolic microenvironment of the bloodstream, where detachment from extracellular matrix induces oxidative stress and anoikis in non transformed cells. Cancer cells counter this by upregulating antioxidant programmes, indicating that blood borne oxygen and nutrient gradients, as well as interactions with circulating immune and stromal cells, configure another host compartment whose properties influence whether disseminated cells can survive transit long enough to seed distant organs. Although detailed survival modeling of time to metastasis is beyond the scope of the available data, these mechanistic insights collectively support a view in which organ specific metabolic and stromal features of the host are central determinants of metastatic colonization patterns [WL21].

3 Genetic and molecular determinants of organ-specific metastasis

3.1 Genomic alterations linked to metastatic competence

Genetic determinants of metastatic competence intersect with both intrinsic tumour biology and host context, shaping organ-specific colonization patterns. Pan cancer comparisons of metabolic gene expression between primary tumours and matched normal tissues show that malignant cells largely preserve the metabolic programme of their tissue of origin, indicating that oncogenic lesions operate within organ-defined metabolic baselines rather than fully overwriting them. Experimental models further reveal that identical oncogenic drivers can imprint distinct metabolic phenotypes depending on tissue site: MYC-driven liver tumours exhibit reduced glutamine anabolism, whereas MYC-driven lung tumours show increased glutamine utilization, and KRAS-based oncogenesis produces divergent branched-chain amino acid metabolism in mouse non small cell lung cancer versus pancreatic cancer. These findings imply that genomic alterations such as MYC or KRAS activation confer metastatic advantage in an organ-dependent manner by co-opting pre-existing tissue metabolic architectures. Context-dependent metabolic routing extends to Ras-driven non small cell lung cancer, where in vivo tumours predominantly use glucose to feed the tricarboxylic acid cycle, yet in vitro counterparts rely on glutamine as the main anaplerotic substrate. This plasticity demonstrates that the same oncogenic configuration can underlie different metabolic phenotypes when nutrient availability shifts, which in turn influences fitness in particular metastatic niches. Host-specific metabolites also modulate the functional impact of genomic lesions. Elevated uric acid levels in human plasma, compared with mice, inhibit uridine monophosphate synthase and reduce sensitivity to 5 fluorouracil, indicating that mutations in pyrimidine biosynthetic enzymes translate into distinct therapeutic vulnerabilities according to species-specific metabolite concentrations [WL21]. Collectively, these observations frame oncogenic mutations as enabling, but not solely dictating, metastatic competence, which emerges from the interaction of driver lesions with organ-level metabolic constraints. Within the brain, lipid-rich conditions create a distinctive selection landscape for genomic programmes that regulate fatty acid and cholesterol metabolism. In triple negative breast cancer models of brain metastasis, RARRES2 is identified as a multifunctional adipokine and chemokine that links lipid metabolic reprogramming to brain colonization. Functional analyses in MDA MB 231 cells show that RARRES2 knockdown decreases PTEN expression and increases phosphorylation of Akt and mTOR, leading to elevated SREBP1 levels and downstream promotion of lipid synthesis. Conversely, RARRES2 overexpression upregulates PTEN and reduces p Akt, p mTOR and SREBP1, positioning RARRES2 as a negative regulator of the PTEN mTOR SREBP1 axis. In patient-derived brain metastasis cases, a negative correlation between RARRES2 and SREBF1 expression corroborates this regulatory relationship in the clinical setting. These data indicate that genomic or epigenetic reduction of RARRES2 expression can drive activation of mTOR dependent lipogenic programmes through PTEN suppression and SREBP1 induction, thereby supporting survival of brain-tropic triple negative breast cancer cells in a lipid-enriched microenvironment. Mechanistic dissection implicates CMKLR1, the primary receptor for RARRES2, as an essential mediator of this signalling axis. CMKLR1 knockdown abolishes RARRES2 overexpression-induced increases in PTEN and decreases in p Akt, and enhances FASN mRNA levels, reinforcing the view that RARRES2 CMKLR1 signalling constrains lipogenic reprogramming. Given that PTEN loss or mutation is reported in approximately 35 to 40% of triple negative breast cancers and ranks among the top genomic alterations in brain metastases from breast cancer, convergence of PTEN inactivation with low RARRES2 expression plausibly amplifies PI3K mTOR SREBP1 pathway activity in this context. Furthermore, prior work associates PI3K pathway mutations with brain metastasis from breast cancer and highlights SREBP1 as selectively required for growth of brain metastatic cells but not for cells lacking brain metastatic potential, emphasizing that genetic lesions in PTEN and PI3K, together with RARRES2 downregulation, configure a lipogenic programme specifically advantageous in the brain. Downstream of these alterations, SREBP1 driven transcription supports de novo synthesis of cholesterol and fatty acids, aligning with the phospholipid and cholesterol dominated lipid composition of the brain. Highly brain metastatic breast cancer cells display increased levels of glycerophospholipid and cholesterol species and decreased triacylglycerols, consistent with adaptation to the low TAG, high glycerophospholipid brain environment and suggestive of selection for genomic configurations that favour SREBP1 dependent glycerophospholipid production [Li+23]. In aggregate, these findings define a genomic circuitry in which RARRES2 and PTEN act as upstream brakes on PI3K mTOR SREBP1 signalling, and their loss or suppression licenses lipid metabolic reprogramming that enhances metastatic competence in the brain.

3.2 Mutations facilitating blood-brain barrier traversal

Evidence on genetic and molecular alterations that specifically facilitate passage of cancer cells across the blood–brain barrier (BBB) is still limited, but several signalling axes implicated in breast cancer brain metastasis intersect with BBB biology and can be considered in this context. Protocadherin 7 (PCDH7) expression on cancer cells promotes formation of carcinoma–astrocyte gap junctions after tumour cells have reached the brain, enabling transfer of cyclic guanosine monophosphate adenosine monophosphate to astrocytes. This intercellular communication triggers production of interferon α and tumour necrosis factor α in astrocytes, which in turn activate STAT1 and NFκB signalling pathways within metastatic cells, enhancing brain colonization and chemoresistance. Although PCDH7 itself is not described as mutated, its aberrant expression establishes a functional interface that lowers the effective barrier posed by BBB-associated astrocytes and converts them into facilitators of metastatic growth [MH21; Far+23]. These findings support the view that genetic programmes controlling junctional adhesion and downstream cytokine signalling are critical modifiers of BBB traversal and post-entry survival. Another molecular system with direct relevance to BBB penetration is the ADAM22–LGI1 axis in endocrine-resistant breast cancer. ADAM22 emerged as a top member of the druggable extracellular matrix gene set in analyses of matched primary and metastatic tumours, including brain lesions, and is highly expressed in breast cancer brain metastases. LGI1 is a specific extracellular ligand for ADAM22 that regulates synaptic transmission through stabilisation of the AMPA/stargazin complex in the nervous system. In cancer models, exogenous LGI1 was previously shown to inhibit migration of endocrine-resistant breast cancer cells. Building on this, a small peptide mimetic (LGI1MIM) was designed against the ligand-binding pocket of LGI1 in the disintegrin domain of ADAM22, and successful binding was confirmed in silico and in vitro. Functionally, LGI1MIM halted migration of aggressive endocrine-resistant cells to levels comparable with endocrine-sensitive MCF7 cells, and in an ADAM22-positive brain metastatic patient-derived model, LGI1MIM reduced proliferation. Importantly, in a 3D biohybrid BBB model containing endothelial cells, astrocytes and patient-derived brain metastatic tumour cells, LGI1MIM and an optimized variant, LGI1MIM-LS, were able to cross the BBB and inhibit tumour cell proliferation. In vivo, LGI1MIM prevented early progression of breast cancer brain metastasis in an endocrine-resistant xenograft model. These data collectively position ADAM22 and its ligand interface as a genetically and structurally defined target whose modulation can influence both BBB traversal and metastatic outgrowth [Cha+20]. Metabolic rewiring also contributes to effective colonization of the brain following BBB passage, and here specific regulatory pathways intersect with genetic lesions. Brain metastatic breast cancer cells adapt to the low-glucose brain milieu by altering gluconeogenesis and enhancing oxidation of glutamine and branched-chain amino acids as alternative energy sources, while using glutamine as a precursor for nucleotides and non-essential amino acids. Overexpression of glucose-regulated chaperone GRP94 coordinates unfolded protein response and pro-survival autophagy under low-glucose conditions, relieving metabolic stress and supporting brain metastases. These adaptive traits, although not tied to discrete point mutations in the available data, underscore that survival after BBB traversal depends on genetically encoded stress-response machinery, including autophagy regulators such as GRP94 and KISS1-mediated survival autophagy, which together help metastatic cells withstand nutrient deprivation and maintain viability in the brain microenvironment [MH21]. Direct genetic changes that remodel lipid metabolism in ways favourable to brain colonization further complement these mechanisms. RARRES2 downregulation in triple negative breast cancer cells reduces PTEN expression and increases phosphorylation of Akt and mTOR, which elevates SREBP1 and drives lipogenic programmes. Given that PTEN loss or mutation is common in triple negative breast cancer and among the top genomic alterations in breast cancer brain metastases, convergence of PTEN inactivation with diminished RARRES2 signalling amplifies PI3K–mTOR–SREBP1 activity. This lipogenic reprogramming enhances de novo synthesis of cholesterol and fatty acids that match the phospholipid- and cholesterol-rich composition of the brain, thereby conferring a survival advantage once cells have crossed the BBB. Highly brain metastatic breast cancer cells exhibit increased glycerophospholipid and cholesterol levels with reduced triacylglycerols, consistent with selection for genomic configurations that favour SREBP1-dependent glycerophospholipid production in the brain niche [Li+23].

3.3 Metabolic mutations conferring survival advantage in the brain

Metabolic rewiring that supports survival in the nutrient and oxygen constrained brain microenvironment arises from genetically encoded alterations that channel carbon flux through specific pathways. Proteomic and metabolomic analyses of brain metastases show that breast cancer cells in the parenchyma increase expression of enzymes in glycolysis, the tricarboxylic acid (TCA) cycle, oxidative phosphorylation (OXPHOS), the pentose phosphate pathway and the glutathione system. This configuration allows simultaneous oxidative metabolism of glucose and robust management of oxidative stress, thereby providing a bioenergetic and redox advantage in a setting characterized by high energy demand and low reserves. ^{13}C-glucose tracing during surgical resection of gliomas and brain metastases further reveals that these tumours oxidize glucose in mitochondria and convert it into lactate, alanine, glutamic acid, glutamine and glycine, consistent with coordinated activation of glycolysis, the TCA cycle, OXPHOS and non-essential amino acid synthesis [WL21]. Although these observations do not map directly to specific mutations, they define a metabolic phenotype in which genetic programmes favor mitochondrial oxidation and ancillary glucose pathways that collectively enhance survival in the brain. Genetic regulation of lipid metabolism contributes a more explicit link between specific alterations and brain-directed fitness. In triple negative breast cancer, RARRES2 functions as an adipokine and chemokine that restrains activation of the PTEN–mTOR–SREBP1 axis. Knockdown of RARRES2 in MDA MB 231 cells decreases PTEN and increases phosphorylation of Akt and mTOR, leading to elevated SREBP1 and downstream promotion of lipid synthesis. Conversely, RARRES2 overexpression upregulates PTEN and reduces *p*-Akt, *p*-mTOR and SREBP1. Patient-derived brain metastasis samples show a negative correlation between RARRES2 and SREBF1 expression, supporting this regulatory relationship in vivo. Given that PTEN loss or mutation occurs in approximately 35 to 40% of triple negative breast cancers and ranks among the most frequent genomic alterations in breast cancer brain metastases, convergence of PTEN inactivation with low RARRES2 expression intensifies PI3K–mTOR–SREBP1 pathway activity. Downstream, SREBP1-driven transcription supports de novo synthesis of cholesterol and fatty acids, aligning with the phospholipid and cholesterol rich composition of brain tissue. Highly brain metastatic breast cancer cells exhibit increased glycerophospholipid and cholesterol levels and reduced triacylglycerols, consistent with selection for genomic configurations that favor SREBP1-dependent glycerophospholipid production and thereby confer a metabolic survival advantage in the brain niche [Li+23]. Lipid handling proteins beyond the RARRES2–PTEN–SREBP1 axis also associate with brain metastasis propensity. Members of the fatty acid binding protein family participate in fatty acid uptake, transport and metabolism. High expression of fatty acid binding protein 7 (FABP7) correlates with poor prognosis and a high incidence of brain metastasis in breast cancer patients. Functional experiments demonstrate that FABP7 promotes adaptation of HER2 positive breast cancer cells to the brain microenvironment by sustaining a glycolytic phenotype and supporting lipid droplet storage. This combination of enhanced glycolysis and fuel buffering through lipid droplets equips metastatic cells to withstand fluctuating nutrient supply in the brain, suggesting that genomic or transcriptional programmes upregulating FABP7 contribute directly to survival advantage after colonization [WL21]. Amino acid metabolism provides additional routes through which genetically regulated pathways improve fitness in the brain. Brain metastatic breast cancer cells can engage gluconeogenesis and branched chain amino acid oxidation to promote survival and growth in the brain microenvironment, and acetate metabolism is also reported to support brain metastases. Although specific mutations are not delineated in these pathways, their activation implies that alterations upstream of gluconeogenic and amino acid catabolic enzymes favor non glucose carbon utilization. In parallel, de novo serine synthesis emerges as a conditionally essential pathway in triple negative breast cancer brain metastasis. Upregulation of 3 phosphoglycerate dehydrogenase (PHGDH), the rate limiting enzyme of glucose derived serine synthesis, is a major determinant of brain colonization, and inhibition of PHGDH attenuates brain metastasis in mice. Limited serine and glycine availability within the brain microenvironment potentiates tumour cell sensitivity to serine synthesis inhibition, indicating that mutations or copy number changes enhancing PHGDH expression or activity would confer a selective advantage in this amino acid poor niche [Ngo+20; WL21]. Stress response pathways intersect with these metabolic programmes to sustain viability under low glucose and oxygen. Overexpression of the glucose regulated chaperone GRP94 in breast cancer brain metastases relieves metabolic stress in low glucose conditions by coordinating the unfolded protein response and pro survival autophagy. KISS1 mediated survival autophagy has also been implicated in brain metastasis, and KISS1 expression differs between primary breast cancers and brain metastases. These findings show that genetically encoded regulation of autophagy and the unfolded protein response enhances tolerance to metabolic stress, complementing lipid, glucose and amino acid pathway alterations that together provide a durable survival advantage within the brain [MH21].

3.4 Molecular determinants of organ-specific colonization beyond the brain

Organ-specific colonization outside the brain is strongly shaped by the metabolic and stromal features of host tissues. Comparative analyses of primary tumours and matched normal organs reveal that cancers largely preserve the metabolic programmes of their tissue of origin, implying that oncogenic lesions act within pre existing organ-defined metabolic baselines rather than fully redefining them. Identical drivers can therefore generate distinct metabolic phenotypes in different organs: MYC-driven liver tumours show reduced glutamine anabolism, whereas MYC-driven lung tumours exhibit increased glutamine utilization, and KRAS-based oncogenesis produces divergent branched-chain amino acid metabolism in mouse non small cell lung cancer versus pancreatic cancer. These findings underscore that the same genomic alterations confer metastatic or survival advantages in an organ-dependent manner by co-opting the specific metabolic architecture of each host tissue. Host metabolism and perfusion further modulate colonization. Ras-driven non small cell lung cancers rely predominantly on glucose to supply the tricarboxylic acid cycle in vivo, yet switch to glutamine as the principal anaplerotic substrate in vitro, demonstrating that nutrient composition in the host circulation dictates pathway usage. Elevated uric acid levels in human plasma, compared with mice, inhibit uridine monophosphate synthase and diminish sensitivity to 5 fluorouracil, so identical genomic lesions in pyrimidine biosynthesis translate into different therapeutic responses depending on species-specific metabolite concentrations. Within single organs, differently perfused regions display distinct metabolic patterns, as shown by multimodality imaging with intraoperative ^{13}C glucose infusions in non small cell lung cancer, while both gliomas and brain metastases in the same brain environment can be fuelled by oxidized glucose and acetate. These examples highlight that systemic and local host metabolism can either restrict or support metastatic colonization according to nutrient and metabolite availability. Secondary sites such as omentum and lymph nodes illustrate how specialized stromal populations generate metabolically supportive niches that favour organ-specific colonization. In the omentum, adipocytes and cancer-associated fibroblasts promote ovarian cancer metastasis by supplying lipids and reprogramming glycogen metabolism. Coculture systems show that activation of p38α MAPK in omentum-derived fibroblasts leads to phosphorylation and activation of glucose phosphate mutase 1 in ovarian cancer cells under normoxia, thereby increasing glycogenolysis and feeding glycolysis to enhance proliferation, invasion and metastatic spread. In vivo, fibroblast-specific depletion of p38α or pharmacologic inhibition of glycogenolysis reduces omental metastasis, demonstrating that host stromal signalling can impose a distinct metabolic orientation on colonizing cancer cells. Lymph nodes constitute another organ context in which molecular determinants of lipid metabolism support colonization. Metastatic melanoma and breast cancer cells in lymph nodes experience a microenvironment enriched in bile acids, which activate YAP via the vitamin D receptor and drive fatty acid oxidation, thereby facilitating adaptation to this lipid-rich niche. In cervical cancer, long noncoding RNA LNMICC recruits NPM1 to the FABP5 promoter, reprogramming fatty acid metabolism and promoting lymph node metastasis through epithelial–mesenchymal transition and VEGF C mediated lymphangiogenesis. Pharmacologic blockade of fatty acid oxidation with etomoxir suppresses lymph node metastasis, indicating that host-derived bile acids, stromal lipid handling and vascular remodelling converge on FAO-dependent programmes that bestow a survival and growth advantage specifically within lymphatic tissue. Circulating tumour cells can be viewed as metastasizing within the circulatory system, where detachment from the extracellular matrix typically induces elevated reactive oxygen species and anoikis in non transformed cells. Cancer cells escape this fate by enhancing intracellular antioxidant programmes, aligning with a broader concept of metabolic adaptation to distinct host compartments. Considered together, these observations support a model in which organ-specific colonization beyond the brain is governed not only by canonical driver mutations but by their integration with organ-level metabolic and stromal cues, particularly those involving glucose, amino acid and lipid metabolism, and by stromal signalling pathways that rewire these metabolic processes in a site-selective manner [WL21]. Nonlinear survival effects and machine learning-based survival models intersect with these biological determinants primarily through their capacity to capture complex interactions among clinical, metastatic and treatment-related variables. In stage IV non small cell lung cancer, Random Survival Forests and XGBoost configured with a Cox loss outperform Cox proportional hazards and single decision tree models in terms of concordance index and integrated Brier score under repeated cross-validation, despite a limited number of events. Hyperparameter optimisation using Bayesian strategies such as Optuna refines these ensemble models by tuning tree number, depth, learning rate and subsampling ratios, whereas Cox and decision tree models are retained with fixed parameters to preserve interpretability. These analyses demonstrate that ensemble machine-learning approaches can more accurately model prognostic relationships in advanced metastatic cohorts, offering a methodological framework that could, in future work, be extended to organ-specific time-to-metastasis or progression endpoints [Als+26].

4 Genes and pathways predisposing to brain metastasis in breast cancer

4.1 Breast cancer subtypes and risk of brain metastases

Intrinsic subtype composition strongly shapes the clinical landscape of breast cancer brain metastases. Gene expression profiling using microarrays and PAM50 classifiers has established at least four major subtypes in primary breast cancer: Luminal A, Luminal B, HER2/ER2, and basal-like, which largely overlaps with triple negative breast cancer (TNBC). In a series of central nervous system metastases from breast cancer, application of the PAM50 classifier showed that many brain lesions retained molecular signatures consistent with their original breast cancer subtype, with a relatively high representation of Luminal B tumours compared with HER2/ER2 and basal-like cases. Although matched primary tumours were not available to confirm subtype conservation, metastatic lesions nonetheless exhibited expression patterns resembling those known for primary breast tumours and their subtypes, implying that intrinsic subtype identity remains a relevant determinant even after colonization of the brain. Clinical and molecular data converge on the observation that HER2/ER2 and basal-like or TNBC subtypes display a particular propensity to seed the brain. These subtypes are reported to have a greater predilection for brain metastasis, shorter latency to CNS involvement, and poorer overall survival than Luminal subtypes. This pattern suggests that subtype-specific biology, including differences in proliferative signalling, invasion capacity and therapy response, contributes to the elevated risk and aggressive course of brain metastases arising from HER2-enriched and basal-like primaries. In the same brain metastasis cohort, deep profiling uncovered fundamental genetic and epigenetic aberrations that operate within these subtype contexts. For example, epigenetic analyses revealed increased DNA methylation overall compared with non-neoplastic tissue, alongside upregulation due to amplification and/or hypomethylation of DNMT3B and MAT1A. Functionally annotated epigenetically regulated genes were strongly related to inflammatory and immunological responses, with a notable tendency for genes involved in tumour and immune cell migration and adhesion to be epigenetically silenced. One proposed explanation is that once cells have colonized the brain, migratory-promoting genes become repressed while proliferative programmes are activated, fitting with the clinically observed rapid progression of brain metastases particularly in aggressive subtypes [Sal+14]. Subtype-specific risk is also intertwined with autophagy and survival pathways that contribute to metastatic fitness in the brain. Triple negative and HER2 breast cancers that metastasize to the brain frequently exhibit amplification of epidermal growth factor receptor (EGFR) accompanied by loss of phosphatase and tensin homolog (PTEN). PTEN acts as a positive regulator of downstream EGFR effectors including PI3K/AKT, and its loss in brain metastases from breast cancer patients is associated with elevated PI3K/AKT signalling that promotes cell survival and c-Myc–driven metabolic reprogramming. These alterations are particularly characteristic of TNBC and HER2 disease, which already show a strong clinical tendency toward brain colonization. Targeting autophagy together with PI3K signalling has therefore been proposed as a potential therapeutic strategy for patients with breast cancer brain metastases arising from these high-risk subtypes [MH21]. Detailed analysis of subtype composition in brain metastases also underscores that these lesions generally retain core features of breast cancer rather than adopting a completely distinct molecular identity. Although receptor conversion between primary and metastatic sites could not be directly assessed in the reported series, gene expression signatures characteristic of primary tumours and their subtypes were identifiable in the CNS lesions. This observation reinforces the need to incorporate intrinsic subtype status when designing and selecting therapeutic approaches for brain metastases, since Luminal, HER2-enriched and basal-like tumours differ markedly in their dominant signalling pathways and expected treatment sensitivity. Because the blood–brain barrier and unique metabolic environment of the brain impose additional constraints, optimal management of brain metastases requires integration of subtype-specific systemic therapies with strategies tailored to the CNS compartment. From a translational perspective, these findings argue that stratification by intrinsic subtype should be central in efforts to develop predictive markers for brain metastasis risk and to guide surveillance in breast cancer patients. HER2/ER2 and basal-like or TNBC subtypes warrant particular attention given their higher propensity for brain involvement and worse survival once metastasis occurs, while Luminal B tumours appear prominently represented among existing brain metastasis series and may also constitute an important subgroup for targeted preventive strategies [Sal+14; MH21].

4.2 Metabolic reprogramming in breast cancer brain metastases

Brain colonization by breast cancer cells is tightly linked to context dependent metabolic rewiring that matches the energetic and biosynthetic constraints of the central nervous system. Proteomic analyses comparing breast cancer brain metastases with extracranial lesions show increased expression of enzymes in glycolysis, the tricarboxylic acid cycle (TCA), oxidative phosphorylation (OXPHOS), the pentose phosphate pathway and the glutathione system. This configuration enables metastatic cells to generate ATP through mitochondrial oxidation of glucose while simultaneously shunting glucose into alternative pathways that support antioxidant defence, thereby maintaining redox homeostasis in an environment characterized by low energy reserves but high energy demand. Infusion of ^{13}C-labelled glucose into patients undergoing resection of gliomas and brain metastases confirms that these tumours oxidize glucose in mitochondria and convert it into lactate, alanine, glutamic acid, glutamine and glycine, consistent with coordinated activation of glycolysis, the TCA cycle, OXPHOS and non-essential amino acid synthesis. These findings delineate a metabolic programme in which parenchymal brain metastases exploit glucose flexibly rather than relying solely on anaerobic glycolysis. Lipid metabolism forms an additional layer of adaptation in breast cancer brain metastases. Members of the fatty acid binding protein family govern fatty acid uptake and intracellular trafficking, and high expression of fatty acid binding protein 7 (FABP7) is associated with poor prognosis and a high incidence of brain metastasis in breast cancer patients. Functional experiments in HER2 breast cancer models demonstrate that FABP7 promotes adaptation to the brain microenvironment by supporting a glycolytic phenotype and facilitating lipid droplet storage. This dual action couples rapid ATP production via glycolysis with fuel buffering in lipid droplets, allowing metastatic cells to withstand fluctuations in nutrient supply within the brain. In parallel, brain metastatic breast cancer cells can invoke gluconeogenesis and branched chain amino acid oxidation to sustain survival and growth, and acetate metabolism has also been implicated in promoting brain metastases, although mechanistic detail remains limited in the available data [WL21]. Amino acid availability in the brain further shapes metabolic reprogramming. Cerebrospinal and interstitial fluids are relatively depleted in most amino acids compared with plasma, with serine and glycine particularly scarce. In this nutrient context, triple negative breast cancer cells that efficiently colonize the brain upregulate de novo serine synthesis. Proteomic comparisons identify 3 phosphoglycerate dehydrogenase (PHGDH), the rate limiting enzyme of glucose derived serine biosynthesis, as the most significantly upregulated protein in brain tropic cells. Genetic silencing or pharmacological inhibition of PHGDH attenuates brain metastasis and prolongs survival in mice, whereas enforced expression of catalytically active PHGDH in non brain tropic cells promotes brain colonization. When cancer cells are grown in tailored media that recapitulate brain interstitial amino acid concentrations, PHGDH dependence is enforced by environmental serine and glycine scarcity, and inhibition of PHGDH selectively impairs proliferation under these brain mimetic conditions. Identical cells cultured in nutrient replete versus brain like media adopt distinct metabolic programmes and drug sensitivities, underscoring that amino acid limitation in the brain unmasking a PHGDH centred vulnerability that would be obscured in standard culture media [Ngo+20]. Glutamine and branched chain amino acids function as alternative carbon sources for brain metastatic breast cancer cells in the low glucose milieu. These cells exhibit altered gluconeogenesis and enhanced oxidation of glutamine and branched chain amino acids, and they use glutamine not only for ATP production via the TCA cycle but also as a precursor to purines, pyrimidines and non essential amino acids. The capacity to channel glutamine into both energy and biosynthesis appears critical for adaptation to the foreign brain environment, where conventional glucose based anabolic pathways are constrained. However, whether heightened glutamine utilization is an intrinsic property of brain metastatic clones or a phenotype acquired during adaptation remains unresolved in current work [MH21]. Stress response pathways interlock with these metabolic changes to confer tolerance to nutrient deprivation. Overexpression of the glucose regulated chaperone GRP94 in breast cancer brain metastases alleviates metabolic stress under low glucose conditions by coordinating the unfolded protein response and pro survival autophagy. KISS1 mediated survival autophagy has also been associated with breast cancer brain metastasis, and reported differences in KISS1 expression between primary breast tumours and brain lesions have led to the proposal that KISS1 driven autophagy could serve both as a diagnostic marker and as a therapeutic axis. More broadly, autophagy regulators such as BECN1, FIP200, p53 and YAP are implicated in breast tumour initiation, progression and metastasis, illustrating that autophagic flux integrates with metabolic rewiring to support survival of disseminated cells within the brain. Manipulating autophagy signalling is therefore being explored as a strategy to disrupt the metabolic compatibility of breast cancer cells with the brain microenvironment [MH21; WL21]. Taken together, research shows that breast cancer brain metastases are characterized by a composite metabolic reprogramming that spans glucose oxidation, alternative amino acid and lipid utilization, and autophagy linked stress responses, with PHGDH dependent serine synthesis, FABP7 supported glycolysis plus lipid storage, and glutamine centred anaplerosis emerging as key components of brain specific metastatic fitness [Ngo+20; WL21; MH21].

4.3 Microenvironmental cooperation in breast cancer brain metastasis

Reciprocal interactions between disseminated tumour cells and brain-resident cells constitute a central axis of cooperation in breast cancer brain metastasis. Astrocytes occupy a particularly prominent position within this network. In the healthy central nervous system they maintain blood–brain barrier (BBB) integrity, deliver nutrients to neurons, regulate inflammatory responses, and shield neurons from hypoxia. These functions rely on astrocytic expression of glucose transporters, which permit glucose influx into the brain, and glycoproteins that export xenobiotics, as well as the capacity to dynamically modulate BBB tightness under stresses such as hypoxia. Breast cancer cells that evade astrocyte mediated elimination subsequently exploit these homeostatic roles, converting astrocytes from guardians of neuronal function into facilitators of metastatic outgrowth. One cooperation route involves reprogramming astrocytes into a reactive state. Breast cancer cells secrete cytokines such as interleukin 1β and CCL2, which promote differentiation of neural progenitor cells into astrocytes and generate reactive astrocytes characterized by upregulation of glial fibrillary acidic protein. These reactive astrocytes release a broad spectrum of neurotrophic and inflammatory factors, including TGFα, ATP, endothelin 1, glutamate, IL 1β, IL 6, TNFα, macrophage inflammatory protein 2, CCL2, CXCL12, glial cell derived neurotrophic factor, nitric oxide and sphingosine 1 phosphate. Collectively, these mediators support metastatic cancer cell growth, progression and therapeutic resistance within the brain parenchyma [MH21]. In this way, tumour cells reshape the cytokine milieu to build a trophic and immunomodulatory niche. Gap junction mediated communication deepens this cooperative relationship. After traversing the BBB, breast cancer cells form gap junctions with astrocytes through the interaction of cancer cell protocadherin 7 and astrocytic connexin 43. Through these junctions, tumour cells transfer $2', 3'$ cyclic GMP AMP into astrocytes, activating the cGAS–STING pathway and driving production of interferon α and TNFα. These cytokines then act back on cancer cells to activate STAT1 and NFκB signalling, enhancing brain metastatic growth and resistance to chemotherapy. Thus, physical coupling between carcinoma and astrocytes establishes a bidirectional signalling loop in which inflammatory pathways in both cell types jointly sustain metastatic fitness. Microglia participate as additional cooperative partners. CD74 positive microglia promote primary brain tumour growth by suppressing antitumour immunity, and in brain metastases, STAT3 positive reactive astrocytes upregulate the CD74 ligand macrophage migration inhibitory factor and increase its binding to CD74 on microglia. This MIF–CD74 interaction activates microglia, leading to upregulation of midkine via NFκB signalling, which promotes development of brain metastases. Cross talk between microglia and reactive astrocytes therefore contributes to an immunosuppressive microenvironment that favours persistence and expansion of metastatic breast cancer cells. In parallel, tumour associated astrocytes upregulate STAT3 and programmed death ligand 1, and secrete immunosuppressive cytokines such as IL 10 and TGFβ, thereby reinforcing local immune evasion [HSA20]. Cancer derived extracellular vesicles (EVs) add an additional layer of long range microenvironmental conditioning. EVs, including exosomes, transport proteins, mRNAs and microRNAs from breast cancer cells to recipient brain cells. They can modulate actin dynamics in endothelial cells, disrupt BBB integrity and facilitate metastatic entry. One study identified exosomes enriched in the Wnt related protein CEMIP derived from breast or lung tumours, which promote brain metastasis by breaking down the BBB and inducing pro inflammatory mediators such as TNF, PTGS2 and CCL/CXCL chemokines in microglia. These changes compromise barrier function and create a pro metastatic inflammatory niche that supports colonization and growth of breast cancer cells in the brain [MH21]. Separately, extracellular vesicles carrying miR 199b from brain tropic breast cancer cells impair metabolic coupling between neurons and astrocytes by downregulating transporters for glutamate, glutamine and lactate. This leads to extracellular retention of glutamine and lactate, which in organotypic brain slice models fuels enhanced growth of metastatic breast cancer cells, illustrating how vesicle mediated reprogramming of neuronal–astrocytic metabolism indirectly supplies nutrients to tumour cells [Rua+24]. Taken together, these findings describe a cooperative microenvironment in which astrocytes, microglia, endothelial cells and neurons are progressively co opted through cytokine signalling, gap junction communication, immunosuppressive programming and EV mediated metabolic rewiring. Rather than acting as a passive scaffold, the brain niche becomes an active participant in breast cancer brain metastasis, with each host cell type contributing specific molecular pathways that collectively enable BBB disruption, immune escape, metabolic support and treatment resistance [MH21; HSA20; Rua+24].

4.4 Therapeutic targets emerging from brain-tropic breast cancer biology

Targetable vulnerabilities in brain-tropic breast cancer emerge at the interface of metabolic rewiring, autophagy regulation and receptor-driven signalling. Autophagy modulation is already under active clinical evaluation in breast cancer, with chloroquine and hydroxychloroquine being tested as autophagy inhibitors in solid tumours. A phase II trial combining the mTOR inhibitor everolimus with hydroxychloroquine reported an increase in six-month median progression-free survival in breast cancer patients, supporting autophagy inhibition as a feasible therapeutic strategy. Additional agents, such as the polyphenolic stilbenoid resveratrol, induce autophagy, inhibit breast cancer stem cell proliferation and suppress Wnt/β-catenin signalling, and can trigger BECN1-independent autophagy-mediated cell death in MCF-7 cells, revealing alternative routes to exploit autophagic machinery for tumour control. Given that KISS1-mediated survival autophagy and GRP94-driven unfolded protein response support breast cancer brain metastases under low-glucose stress, these clinically tractable autophagy modifiers provide a foundation for targeting survival pathways that are particularly relevant in the brain milieu [MH21]. Metabolic adaptation pathways constitute another major source of therapeutic targets. In brain metastases from breast cancer, high expression of FABP7 promotes adaptation of HER2-positive cells by sustaining a glycolytic phenotype and lipid droplet storage, while brain metastatic cells employ gluconeogenesis and branched-chain amino acid oxidation to survive in the nutrient-constrained parenchyma. These observations nominate fatty acid handling and associated oxidative pathways as intervention points. OXPHOS inhibitors have already been shown to improve survival in intracranial melanoma xenograft models and to suppress spontaneous brain metastasis of melanoma, indicating that mitochondrial metabolism can be pharmacologically constrained in brain lesions. In parallel, inhibition of CD36 or modulation of SPARC-driven interactions between ovarian cancer cells and omental adipocytes effectively blocks lipid-fuelled omental metastasis, illustrating that targeting lipid uptake and adipocyte–tumour metabolic coupling can curtail metastasis in lipid-rich niches. By analogy, disrupting lipid uptake or fatty acid oxidation in FABP7-high, brain-tropic breast cancers represents a rational avenue to impair their metabolic compatibility with the brain microenvironment [WL21]. Ferroptosis-based strategies offer a complementary line of attack for metastatic breast cancer. Neratinib, a pan-HER tyrosine kinase inhibitor that binds irreversibly to HER1, HER2 and HER4, induces ferroptotic cell death in HER2-positive models such as TBCP-1 and SKBR3. Treatment with ferroptosis inducers like erastin or RSL3 kills these cells, and liproxstatin-1, a ferroptosis inhibitor, rescues them, confirming on-target ferroptotic death. Importantly, neratinib-induced cell death is prevented by liproxstatin-1, demonstrating that neratinib acts in part by triggering ferroptosis, whereas other TKIs including erlotinib, lapatinib, tucatinib and afatinib do not show liproxstatin-1-reversible effects. These data identify a HER2-directed agent with a unique ferroptotic mechanism in breast cancer [Nag+19]. Broader analyses of ferroptosis-related genes in breast cancer metastasis further link altered iron metabolism to prognosis and nominate small-molecule modulators derived from connectivity mapping as potential therapeutics, underscoring ferroptosis as an emerging axis in metastatic control [Zhu+22]. Molecularly targeted therapies directed at HER family receptors and downstream pathways are already in clinical use for breast cancer brain metastases. Combination regimens of novel anti-HER2 agents plus capecitabine have shown efficacy in patients with HER2-positive disease and established brain metastases. Neratinib plus capecitabine achieved a central nervous system response rate of 49%, compared with 8% for neratinib alone, while addition of the HER2-selective TKI tucatinib to trastuzumab and capecitabine improved CNS response and progression-free survival in heavily pretreated HER2-positive patients. Cyclin-dependent kinase 4/6 inhibition with abemaciclib plus endocrine therapy has reduced metastatic tumour load in a substantial fraction of hormone receptor-positive, HER2-negative patients with brain metastases, extending targeted approaches beyond HER2-amplified disease. At the level of microenvironmental signalling, inhibition of STAT3 with silibinin, which crosses the blood–brain barrier, impairs viability of brain metastases in both mice and humans, likely via effects on tumour-associated astrocytes and their cross talk with cancer cells and microglia. The JAK inhibitor ruxolitinib similarly limits growth of primary brain tumours and reduces

STAT3-positive reactive astrocytes in mice. Gap junction modulators such as meclofenamate and tonabersat, which disrupt carcinoma–astrocyte communication, inhibit brain metastatic outgrowth, highlighting intercellular conduits as druggable structures in the brain niche [HSA20]. Collectively, these interventions converge on autophagy, lipid and mitochondrial metabolism, ferroptosis and HER–STAT3–gap junction signalling as actionable dimensions of brain-tropic breast cancer biology, with several agents already in clinical or preclinical testing in relevant metastatic settings [MH21; WL21; Nag+19; HSA20].

5 Organ-specific metastasis in lung cancer and other primaries

5.1 Brain metastases in non-small cell lung cancer

Stage IV non small cell lung cancer (NSCLC) is characterized by poor survival, with most patients diagnosed at an advanced or metastasized stage when treatment is largely palliative and long term survival is uncommon. Approximately 85% of lung cancer cases fall into the NSCLC category, and outcomes are shaped by a multifactorial interplay of demographic, clinical, pathological and molecular variables, which makes accurate prognostic modeling essential for advanced disease management. Within this context, brain involvement represents one of several metastatic sites contributing to survival heterogeneity, although the available data focus on overall advanced stage rather than isolating brain metastases as a distinct endpoint. Methodological work in advanced NSCLC provides detailed insight into time to event modeling strategies that can, in principle, accommodate nonlinear survival effects associated with organ specific metastases. The Cox proportional hazards model has long been the benchmark because of its interpretability and capacity to handle censored outcomes. It assumes a multiplicative hazard structure with time invariant hazard ratios, as reflected in the formulation of the hazard function and hazard ratio based on covariate vectors and regression coefficients. This proportional hazards assumption was formally checked using scaled Schoenfeld residuals and found to be adequately satisfied in the analyzed stage IV lung cancer cohort. Nevertheless, these assumptions are frequently violated in complex, high dimensional cancer datasets, which motivates the use of machine learning based survival methods. Random Survival Forests (RSF) and XGBoost configured with a Cox loss have been applied to stage IV lung cancer data to capture nonlinearities and higher order interactions among clinical, metastatic and treatment related covariates. Using repeated cross validation in a single institution cohort with $n = 47$ observed events, RSF achieved the highest concordance index (0.748 ± 0.036) and the lowest Integrated Brier Score (IBS $= 0.118$), indicating strong discrimination and calibration. XGBoost displayed comparable performance (C index $= 0.732 \pm 0.041$, IBS $= 0.124$), outperforming the Cox model (C index $= 0.718 \pm 0.044$, IBS $= 0.141$) and a single decision tree (C index $= 0.658 \pm 0.069$, IBS $= 0.178$). These results underscore the superiority of ensemble machine learning methods in modeling intricate multidimensional prognostic links in advanced lung cancer. Hyperparameter optimization using Bayesian strategies implemented via Optuna was central to achieving these gains. For RSF, tuning targeted the number of trees (300 to 1000), minimum leaf size, and feature sampling strategy, yielding an optimal configuration with 642 trees, a minimum leaf size of 8, and $\log 2$ feature sampling. For XGBoost, the optimized parameters included 735 estimators, learning rate 0.1402, maximum depth 3, and subsample ratio 0.7811. In contrast, Cox and decision tree models were retained with fixed parameter settings to preserve interpretability and allow fair comparison. Model reliability assessment using IBS confirmed that RSF and XGBoost provided more accurate probabilistic survival predictions across the follow up period, supporting the use of ensemble machine learning approaches over traditional regression alone in advanced stage NSCLC. Clinical covariates related to metastatic burden emerged as recurrent prognostic markers in these analyses. Pleural metastasis and liver metastasis, together with radiation therapy, consistently appeared as important features across Cox, decision tree, RSF and XGBoost models. Kaplan–Meier risk stratification based on model derived risk scores demonstrated clear separation between high risk and low risk patient cohorts, reinforcing the clinical relevance of these covariates. While specific brain metastasis variables were not individually highlighted, the alignment of feature importance across methodological frameworks suggests that particular metastatic locations can exert strong influence on survival when appropriately modeled. Several limitations temper direct extrapolation of these findings to brain metastases in NSCLC. The study relied on a modest, single institution dataset without external validation, which constrains statistical power for assessing the independent impact of individual metastatic sites and molecular biomarkers.

Potential confounders such as prior treatment history, comorbidities, detailed radiologic characteristics and socioeconomic factors were not captured. Ensemble models, despite superior predictive performance, remained relatively opaque, and comprehensive explainability analyses such as SHAP values or partial dependence plots were beyond the study scope. Temporal changes in treatment modalities and supportive care also introduced potential bias. Future multi center studies with richer clinical and molecular annotation are advocated to strengthen generalizability and to link advanced survival modeling, including RSF and XGBoost, more explicitly to organ specific metastatic patterns such as brain involvement in NSCLC [Als+26].

5.2 Organotropism in lung cancer with emphasis on brain involvement

Patterns of organotropism in lung cancer can be partially inferred from survival modeling studies that integrate metastatic site information into time to event analyses. In advanced non small cell lung cancer (NSCLC), prognostic modeling based on stage IV cohorts underscores the importance of metastatic burden variables among the determinants of overall survival. A Cox proportional hazards model applied to such a cohort showed adequate fulfillment of the proportional hazards assumption when assessed using scaled Schoenfeld residuals, supporting its use as a baseline framework for time dependent outcome analysis in this setting. Nevertheless, the same dataset was used to benchmark more flexible approaches, revealing organ related covariates as consistent contributors to risk stratification. Comparative evaluation of Random Survival Forest (RSF), XGBoost configured with a Cox loss, Cox proportional hazards, and a single decision tree model demonstrates that ensemble methods achieve superior predictive performance for stage IV lung cancer survival. Using repeated cross validation in a cohort with $n = 47$ events, RSF attained a mean concordance index of 0.748 ± 0.036 and an Integrated Brier Score (IBS) of 0.118, whereas XGBoost yielded a concordance index of 0.732 ± 0.041 and IBS of 0.124. In contrast, the Cox model reached a concordance index of 0.718 ± 0.044 with IBS 0.141, and the decision tree performed worst with a concordance index of 0.658 ± 0.069 and IBS 0.178. These results highlight that ensemble based machine learning methods better capture multidimensional prognostic relationships in advanced lung cancer than traditional Cox regression or single tree models. Hyperparameter optimization using Bayesian strategies implemented via Optuna contributed to the robustness of the ensemble models. For RSF, the number of trees, minimum leaf size and feature sampling strategy were tuned within predefined ranges, resulting in an optimal configuration of 642 trees, minimum leaf size 8 and $\log 2$ feature sampling. For XGBoost, tuning targeted the number of estimators, learning rate, maximum depth and subsampling ratio, with final values of 735 estimators, learning rate 0.1402, depth 3 and subsample 0.7811. Cox and decision tree models were intentionally kept with fixed parameter settings to preserve interpretability and facilitate methodological comparison. Integrated Brier Score based reliability assessment confirmed that RSF and XGBoost provided the lowest prediction errors, supporting their use for probabilistic survival prediction in advanced stage lung cancer. Within this modeling framework, specific metastatic sites emerge repeatedly as influential predictors. Pleural metastasis and liver metastasis, together with receipt of radiation therapy, consistently appear as important prognostic markers across Cox, decision tree, RSF and XGBoost models. Kaplan–Meier curves based on model derived risk scores show clear separation of high risk and low risk groups, reinforcing the clinical relevance of these covariates. Although brain metastasis is not singled out as an individual variable in the available data, the analytical strategy demonstrates how organ specific involvement can be incorporated into multivariable survival models and how ensemble methods can detect nonlinear or interaction effects between metastatic sites and other clinical factors. Methodological considerations temper direct conclusions about brain directed organotropism from this single institution cohort. Sample size is modest and limited to a single center, reducing statistical power to estimate independent effects of separate metastatic locations and molecular biomarkers. External validation is lacking, so generalizability to other populations and treatment eras remains uncertain. Potential confounders, such as prior therapies, comorbidities, detailed radiologic features and socioeconomic variables, were not captured. Moreover, RSF and XGBoost models, though superior in discrimination and calibration, are relatively opaque; comprehensive explainability analyses using SHAP values or partial dependence plots were not undertaken. Nonetheless, the alignment of key prognostic indicators across statistical and machine learning frameworks supports the reliability of the identified metastatic and treatment related predictors and motivates future multi center studies to extend these approaches to brain specific outcomes in NSCLC. From a broader perspective, this work illustrates how organotropism in lung cancer, including brain involvement, can be approached quantitatively through survival modeling rather than purely descriptive clinical observations. Ensemble based survival models such as RSF and XGBoost offer a practical means to integrate metastatic site, treatment variables and host factors into a unified prognostic system, providing a methodological platform that could be adapted to analyze brain metastases explicitly once appropriately annotated datasets become available [Als+26].

5.3 Comparative organ-specific metastasis patterns in other cancers

Comparisons across tumour types highlight both shared and distinct features of organ-specific metastatic patterns that extend beyond lung and breast primaries. One axis concerns how host organ metabolism and perfusion constrain or enable colonization. Pan-cancer analyses of metabolic gene expression show that primary tumours largely retain the metabolic programme of their tissue of origin, indicating that oncogenic drivers act within organ-defined metabolic baselines rather than fully remodelling them. Ras-driven non-small-cell lung cancers (NSCLC) provide a clear example of environmental dependence: in vivo these tumours mainly use glucose to feed the tricarboxylic acid (TCA) cycle, whereas in culture glutamine becomes the principal anaplerotic substrate. Differences in host physiology further modulate metastatic behaviour; uric acid levels in mouse plasma are approximately one-tenth of those in humans, such that uridine monophosphate synthase inhibition and 5-fluorouracil resistance driven by uric acid in human tumours are largely absent in mouse models. Within individual organs, perfusion heterogeneity produces intra-tumoural metabolic diversity, as shown by multimodality imaging with intraoperative ^{13}C-glucose infusions in NSCLC. In the central nervous system, both gliomas and brain metastases in the same environment can rely on oxidized glucose and acetate without glutamine oxidation, underscoring that distinct primaries converging on the brain may adopt similar metabolic solutions [WL21]. Metastatic adaptation to nutrient stress is not confined to breast cancer. NSCLC cells subjected to glutamine deprivation reprogramme amino acid metabolism via a mutant KRAS–Akt–NRF2 axis. Activation of NRF2 upregulates activating transcription factor 4, which in turn increases transcription of asparagine synthetase and promotes asparagine synthesis, while also enhancing uptake of extracellular amino acids to replenish pools depleted by glutamine scarcity. When glutaminase is inhibited, NSCLC cells compensate by upregulating glutamic pyruvic transaminase 2, enabling maintenance of glutamate and pyruvate production and thereby preserving glutamine availability despite reduced input [Pei+25]. Colorectal cancer derived SW620 cells illustrate a related but distinct adaptive profile. Under inhibition of oxidative phosphorylation (OXPHOS) with oligomycin A, intracellular ATP levels in SW620 cells initially rise above basal values before stabilizing near baseline, suggesting engagement of compensatory metabolic pathways that buffer energy supply more effectively than in their primary tumour counterparts SW480. SW620 cells also show stronger OXPHOS and glycolytic potential and higher expression of glycolytic regulators such as HK1, HK2 and GLUT1 under glucose replete conditions, indicating a broader metabolic repertoire that can support survival in diverse secondary niches [Che+18]. Beyond metabolic plasticity, circulating tumour cells (CTCs) constitute a distinct metastasis-related compartment in which survival depends on adaptation to the circulatory microenvironment. Detachment from the extracellular matrix typically induces oxidative stress and anoikis in non-transformed cells, whereas CTCs from breast cancer patients show enhanced antioxidant programmes that permit persistence in the bloodstream. Research on CTCs has focused on antioxidant metabolism, but conceptual frameworks propose treating CTCs as a form of circulatory system metastasis to explore how they respond to heterogeneity in arterial and venous oxygen or nutrient content and to metabolic interactions with other blood-borne cells [WL21]. Experimental isolation of CTCs from breast cancer patients using lineage depletion and positive selection for PanCK or CD44/CD24 phenotypes confirms their tumour origin: these cells are absent in healthy donors and harbour hotspot TP53 mutations not present in matched CD45/CD34 normal blood cells, reinforcing that genetically altered clones circulate and may seed distant organs [Bor+17]. Comparative survival modeling across cancers complements these biological insights by quantifying how organ-specific metastatic patterns contribute to prognosis. In stage IV lung cancer cohorts, Cox proportional hazards models remain a reference owing to interpretability, yet ensemble-based machine learning methods such as Random Survival Forests (RSF) and XGBoost configured with a Cox loss consistently show superior predictive accuracy. RSF attains higher concordance indices and lower Integrated Brier Scores than Cox and single decision tree models under repeated cross-validation, indicating better discrimination and calibration in high-dimensional settings. Hyperparameter optimization with Bayesian strategies implemented via Optuna tunes parameters including tree number, depth, learning rate and subsampling to achieve stable performance. Model reliability assessment via Integrated Brier Score confirms lower prediction error for RSF and XGBoost, supporting ensemble approaches for probabilistic survival prediction in advanced-stage lung cancer. Feature importance analyses identify pleural and liver metastases, along with radiation therapy and demographic factors such as age and body mass index, as recurrent prognostic markers, underscoring that specific metastatic sites exert measurable survival effects when modeled appropriately [Als+26]. Although these studies focus on overall survival rather than time to metastasis and use Cox-based losses rather than accelerated failure time formulations, they illustrate how machine learning survival models can capture complex, potentially nonlinear relationships between organ involvement, treatment, and outcome across cancer types.

6 Quantification and modeling of brain metastasis risk

6.1 Traditional survival analysis methods in oncology

Cox proportional hazards regression occupies a central position among traditional survival analysis tools used in oncology. It is applied as a multiple regression method for time to event data, allowing assessment of how covariates influence individual hazard functions while leaving the baseline hazard unspecified. In this semi parametric framework, the hazard at time t for an individual with covariate vector Y is modeled as

$$\mathcal{L}(t \mid Y, \boldsymbol{\beta}) = \Pi_0(t)\exp(Y\boldsymbol{\beta}), \tag{1}$$

where $\Pi_0(t)$ denotes the baseline hazard and $\boldsymbol{\beta}$ is the vector of regression coefficients. The ratio of hazards for two subjects with covariate vectors Y_1 and Y_2 is

$$\mathrm{HR}(t \mid Y_1, Y_2) = \exp\big(\boldsymbol{\beta}(Y_2 - Y_1)\big), \tag{2}$$

which is time invariant because of the multiplicative structure, embodying the proportional hazards assumption. This property, together with the ability to handle censoring, underpins the widespread use of Cox models in oncologic survival analyses. Applications in advanced non small cell lung cancer (NSCLC) illustrate both strengths and boundaries of this traditional approach. In a stage IV cohort, Cox proportional hazards modeling served as the benchmark for comparing prognostic methods. Preprocessing and model construction followed a reproducible pipeline, and assessment of the proportional hazards assumption using scaled Schoenfeld residuals showed no significant violations, indicating adequate conformity to model assumptions in that dataset. Nonetheless, the analysis explicitly acknowledges that in high dimensional cancer settings these assumptions are often at risk of being violated, which can degrade discrimination and calibration. This motivates careful diagnostic checking whenever Cox models are used to evaluate survival in complex metastatic disease. Within that lung cancer study, Cox modeling delivered moderate predictive performance, achieving a concordance index of 0.718 ± 0.044 and an Integrated Brier Score (IBS) of 0.141 under repeated cross validation. These metrics quantify discrimination and time dependent prediction error, respectively, and provide a reference for evaluating alternative methods. Importantly, Cox regression retained interpretability advantages: regression coefficients in $\boldsymbol{\beta}$ map directly to hazard ratios via equation 2, allowing clinicians to relate covariate changes such as pleural or liver metastasis presence to proportional shifts in hazard. Feature importance analyses in that framework consistently identified pleural metastasis, liver metastasis and radiation therapy as key prognostic markers, demonstrating that even when more flexible methods outperform Cox in accuracy, traditional models still yield clinically meaningful insights into organ specific metastatic burden and treatment effects [Als+26]. Traditional Cox based strategies also extend beyond purely methodological expositions into concrete trial analyses. In a cohort of breast cancer patients with brain metastases, overall survival after brain involvement was modeled using Cox regression, with hazard ratios and 95% confidence intervals estimated alongside Kaplan–Meier curves for brain metastasis overall survival. Variables identified as significant in univariate testing were entered into multivariable Cox models, and independent prognostic factors were selected via backward stepwise procedures. A nomogram was then constructed directly on this Cox backbone, with Harrell's C index used to quantify discriminative ability and bootstrap resampling employed for internal validation. This workflow exemplifies how Cox regression underpins clinically oriented tools that estimate survival probabilities for patients with breast cancer brain metastases, thereby informing risk stratification and treatment planning [Hua+18]. Cox based analyses further appear in trials of systemic and local therapies for brain metastatic breast cancer. Progression free survival and overall survival have been defined as time from treatment initiation to progression or death, and estimated using Cox models with hazard ratios and confidence intervals, supported by Kaplan–Meier plots. Such analyses often incorporate corrections for multiple comparisons, for example through Holm–Bonferroni procedures, underscoring the integration of classical statistical rigor with time to event endpoints in metastatic settings [Leo+20]. Despite these advantages, comparative work in stage IV NSCLC documents that ensemble machine learning methods configured for survival, such as Random Survival Forests and XGBoost with a Cox loss, can surpass Cox regression in concordance index and IBS. In that cohort, Random Survival Forest achieved a C index of 0.748 ± 0.036 and IBS of 0.118, while XGBoost reached 0.732 ± 0.041 and 0.124, respectively, outperforming Cox on both discrimination and calibration. Nonetheless, the study emphasizes that decision trees and Cox models preserve transparency, and advocates hybrid analytical strategies that combine performance oriented machine learning with interpretable regression. This position reflects an emerging consensus that traditional Cox models remain foundational in oncology survival analysis, particularly when clear explanation of covariate effects is essential, even as more flexible approaches gain prominence for complex metastatic prognostic tasks [Als+26].

6.2 Nomograms and clinical risk scores

Clinical risk scores and nomograms are commonly constructed on a Cox proportional hazards backbone, translating multivariable regression outputs into individualized survival predictions. Within metastatic oncology, such tools have been applied directly to patients with lung primaries as well as those with brain involvement from breast cancer, offering structured frameworks for prognostic estimation and treatment planning. One example comes from a large cohort of lung cancer patients, in which age, marital status, gender, race, tumour grade, T stage and primary tumour site were evaluated as potential prognostic factors. Significant variables from univariate analyses were entered into a multivariable Cox proportional risk model, yielding independent prognostic covariates with hazard ratios and 95% confidence intervals. Protective factors included being married, female sex, white race and specific primary sites, whereas higher age, advanced grade and advanced T stage acted as risk factors for decreased overall survival. Based on these multivariable coefficients, a nomogram was constructed that assigns weighted scores to each covariate, sums them to a total score, and maps this score to predicted overall survival probabilities. Internal validation relied on discrimination and calibration: the concordance index (C index) quantified the ability of the model to distinguish patients with different survival times, and calibration plots compared observed versus nomogram predicted overall survival, demonstrating correlation between predicted and actual outcomes [Yua+22]. Comparable methodology has been applied in metastatic breast cancer with brain involvement. In a cohort of patients with breast cancer brain metastases, overall survival after brain metastasis was modeled using Cox regression. Variables significant in univariate analyses were incorporated into multivariable models, and backward stepwise selection was used to identify independent prognostic factors. A nomogram was then developed on this Cox framework to estimate brain metastasis overall survival. Harrell's C index served as the primary metric of discriminative ability, and bootstrap resampling underpinned internal validation, mirroring the lung cancer study's approach to calibration. This workflow illustrates how brain metastasis specific covariates can be integrated into clinically usable risk scoring tools, providing individualized survival estimates that can inform decisions about systemic therapy, local brain-directed treatment and follow up intensity [Hua+18]. Beyond nomogram construction, time to event endpoints in metastatic studies frequently rely on classical survival analysis. For example, in a clinical trial evaluating pyrotinib based regimens in HER2 positive metastatic breast cancer, Kaplan–Meier plots of progression free survival were generated and log rank tests applied to compare subgroups stratified by metastatic site number or prior tyrosine kinase inhibitor exposure. Although no formal nomogram was built, this analysis exemplifies the routine use of risk stratification curves alongside hazard based methods to interpret treatment efficacy in patients with brain metastases, as one complete response occurred in a patient with brain metastasis receiving pyrotinib plus capecitabine and whole brain radiotherapy as first line therapy [Lin+20]. Contemporary comparisons between Cox based models and machine learning frameworks highlight both the persistence of nomogram style risk scoring and the emergence of ensemble methods. In stage IV non small cell lung cancer, Cox proportional hazards regression was used as the benchmark, with proportional hazards assumptions checked via Schoenfeld residuals and found to be adequately satisfied. Covariate effects were interpreted through hazard ratios derived from regression coefficients, maintaining transparency that is central for nomogram construction. However, Random

Survival Forests and XGBoost configured with a Cox loss demonstrated higher concordance indices and lower Integrated Brier Scores than Cox and single decision tree models under repeated cross validation, indicating improved discrimination and calibration. Hyperparameter optimization with Bayesian strategies such as Optuna tuned key parameters including tree number, depth and learning rate for the ensemble models, while Cox parameters were kept fixed to preserve interpretability. Conclusions from this work explicitly state that, despite the superior predictive performance of RSF and XGBoost, transparent models such as decision trees and Cox proportional hazards remain critical, and a hybrid analytical strategy combining performance oriented machine learning with interpretable regression is advocated. This view aligns with the continued reliance on Cox based nomograms and clinical risk scores, especially where clear explanation of covariate effects for brain metastasis risk and survival is required [Als+26; Yua+22; Hua+18; Lin+20].

6.3 Nonlinear and machine learning based survival models

Nonlinear survival behavior in metastatic oncology has motivated the adoption of modeling frameworks that move beyond strictly linear, proportional hazards assumptions. In stage IV non small cell lung cancer, Cox proportional hazards regression has historically served as the benchmark owing to its interpretability and capacity to accommodate censored outcomes, yet key assumptions such as proportional hazards and linear covariate effects are often at risk in high dimensional settings. Formal assessment of the proportional hazards assumption with scaled Schoenfeld residuals in one advanced lung cancer cohort revealed no major violations, but the authors explicitly noted that in complex cancer data these assumptions can be extensively breached, which degrades discrimination and calibration. This recognition has catalyzed systematic comparisons between Cox-based approaches and more flexible survival models capable of capturing nonlinearities and higher order interactions. Machine learning based survival methods, particularly Random Survival Forests (RSF) and gradient boosting variants such as XGBoost configured with a Cox loss, have been applied to stage IV lung cancer to address these limitations. In a carefully tuned single institution cohort, RSF achieved the highest concordance index, 0.748 ± 0.036, together with the lowest Integrated Brier Score (IBS = 0.118), indicating superior discrimination and lower time dependent prediction error relative to alternative models. XGBoost showed comparable performance, with a concordance index of 0.732 ± 0.041 and IBS of 0.124. By contrast, the Cox proportional hazards model reached a concordance index of 0.718 ± 0.044 and IBS of 0.141, while a single decision tree performed worst (C index = 0.658 ± 0.069, IBS = 0.178). These results underscore the noticeable superiority of ensemble machine learning methods in modeling intricate multidimensional prognostic relationships in advanced cancer cohorts. Methodological analysis in this lung cancer study details how hyperparameter optimization contributes to these gains. Bayesian optimization via Optuna was used to tune RSF hyperparameters, including the number of trees, feature sampling strategy and minimum leaf size, yielding an optimal configuration of 642 trees, minimum leaf size 8 and $\log 2$ feature sampling. For XGBoost with a Cox loss, tuning focused on tree depth, learning rate, number of estimators and subsampling ratio, with selected values of 735 estimators, learning rate 0.1402, maximum depth 3 and subsample 0.7811. In contrast, Cox and decision tree models were intentionally retained with fixed parameter settings to preserve interpretability and ensure fair comparison. Reliability assessment with IBS confirmed that RSF and XGBoost provided the most accurate probabilistic survival predictions over the follow up period, reinforcing the utility of ensemble based machine learning for time to event modeling in advanced lung cancer. Beyond this specific cohort, research on survival modeling in oncology highlights two further points relevant for quantifying metastasis risk. First, comparative studies in early stage breast cancer and stage III colorectal cancer reported only modest improvements in concordance indices for machine learning models relative to Cox, and frequently emphasized that many analyses concentrated on discrimination metrics such as C index and area under the curve while underreporting calibration, Brier scores or net clinical benefit. This pattern raises concerns that even when machine learning yields incremental accuracy gains, clinical utility and generalizability may remain underexplored if evaluation is restricted to discrimination alone. Second, the same lung cancer study articulated a structured framework that integrates reproducible preprocessing and tuning, expanded evaluation metrics including calibration and clinical usefulness, feature importance analyses, and robust internal validation through repeated cross validation. This framework was explicitly designed to reconcile methodological rigor with clinical relevance, enabling interpretable, actionable prognostic models despite the use of complex machine learning techniques. With respect to the specific interest in accelerated failure time (AFT) models and nonlinear survival effects, the available sources mention AFT in the context of prior comparative work but do not provide technical details on AFT implementations or their application to metastasis, recurrence or progression endpoints. Similarly, while XGBoost is clearly used in a Cox loss configuration for survival in stage IV lung cancer, there is no explicit description of XGBoost combined with AFT distributions or of models directly targeting time to metastasis or brain specific progression. The current evidence therefore supports a conclusion that ensemble based machine learning survival models, particularly RSF and XGBoost with Cox loss, can outperform traditional Cox regression in advanced lung cancer and that carefully tuned, well validated pipelines are essential for exploiting their advantages, but it does not yet document AFT based or metastasis specific implementations of these methods [Als+26].

6.4 Accelerated failure time and other parametric models

Accelerated failure time (AFT) and other fully parametric survival models are conceptually attractive for metastasis research because they can describe nonlinear covariate effects on event times, but explicit applications in organ specific metastasis, including brain involvement, remain scarce within the available evidence. Comparative survival modeling in oncology has instead centered predominantly on Cox proportional hazards (CoxPH), with AFT often mentioned only as a comparator rather than a primary framework. For example, one analysis in early stage breast cancer evaluated AFT, Random Survival Forests and Gradient Boosting against CoxPH, reporting only slightly higher concordance indices for machine learning models relative to Cox, yet provided limited detail on the specific AFT distributions or parameterizations used and did not extend outcomes to metastasis, recurrence or progression. Similar work in stage III colorectal cancer implemented a deep learning Cox hybrid model with C indices between 0.71 and 0.74, illustrating interest in nonlinear effects but again favoring Cox based formulations over explicit AFT structures, and focusing on discrimination rather than calibration, Brier score or decision curve analysis. In advanced non small cell lung cancer, parametric approaches such as AFT are acknowledged as part of the broader methodological landscape, yet the detailed modeling work relies on semi parametric and machine learning methods that retain Cox style hazard formulations. A stage IV lung cancer study was explicitly designed to benchmark CoxPH against decision trees, Random Survival Forests (RSF) and XGBoost configured with a Cox loss on time to death, rather than on time to metastasis or organ specific progression. Computational methods sections describe CoxPH in standard form, with the hazard $\mathcal{L}(t \mid Y, \boldsymbol{\beta})$ expressed as a product of a baseline hazard $\Pi_0(t)$ and an exponential term in the covariates, and emphasize the proportional hazards assumption, assessed via scaled Schoenfeld residuals. Decision tree, RSF and XGBoost implementations then build on this Cox backbone, for example by using a Cox type loss in XGBoost and by treating RSF as a survival extension of Random Forests, but no parametric baseline distributions or log time models characteristic of AFT formulations are introduced. Performance comparisons in that stage IV cohort are therefore framed entirely in terms of Cox aligned metrics such as hazard ratios, concordance index and Integrated Brier Score (IBS). RSF achieved the highest mean concordance index, 0.748 ± 0.036, and the lowest IBS, 0.118, followed by XGBoost with a Cox survival objective (C index 0.732 ± 0.041, IBS 0.124), whereas CoxPH obtained a concordance index of 0.718 ± 0.044 and IBS of 0.141, and a single decision tree performed worst. Hyperparameter optimization via Bayesian strategies implemented in Optuna tuned RSF parameters such as number of trees, minimum leaf size and feature sampling, and XGBoost parameters including number of estimators, learning rate, depth and subsample ratio, while CoxPH remained with fixed penalization for interpretability. These choices underscore a methodological preference for enhancing nonlinear modeling capacity through tree based ensembles with Cox losses, rather than through parametric AFT specifications of survival time distributions [Als+26]. Research on pyrotinib based therapy in HER2 positive metastatic breast cancer, including patients with brain metastases, further exemplifies reliance on traditional non parametric and Cox tools rather than AFT or alternative parametric models. Progression free survival and overall survival were defined as times from treatment initiation to progression or death and were estimated using the Kaplan–Meier method, with subgroup comparisons assessed via log rank tests. Median follow up was derived using reverse Kaplan–Meier, and stepwise Cox regression was applied to identify factors associated with progression free survival. No parametric distributional assumptions beyond the semi parametric Cox structure are reported, and there is no indication that AFT formulations were employed to capture potential nonlinear time scaling effects of covariates such as brain metastasis status or prior HER2 targeted therapy exposure [Lin+20]. Similarly, nomogram based prognostic tools for breast cancer brain metastases and for lung cancer cohorts have been constructed on multivariable Cox models, with discrimination assessed through Harrell's C index and calibration evaluated via bootstrap resampling and calibration plots. These studies do not describe the use of AFT distributions or other parametric survival functions in model building, nor do they report parametric estimates of time ratios that are characteristic outputs of AFT analysis. Instead, they reinforce the dominance of Cox based approaches as the statistical backbone for clinical prognostic instruments, even when more flexible machine learning models are explored in parallel [Hua+18; Yua+22; Als+26]. Taken together, available evidence points to a methodological gap: while AFT models are recognized in comparative discussions and nonlinear survival behavior is an explicit motivation for adopting alternative techniques, concrete implementations of AFT or other fully parametric survival models for metastasis related endpoints, including brain metastasis risk, are not detailed. Likewise, applications combining XGBoost with AFT distributions for time to event outcomes such as time to metastasis, recurrence or progression are not documented. Instead, ensemble machine learning for survival has so far been realized predominantly through RSF and XGBoost configured with Cox type losses applied to overall survival in advanced lung cancer and progression free survival in metastatic breast cancer [Als+26; Lin+20].

7 Advanced time-to-event modeling for metastasis, recurrence, and progression

7.1 Applications of gradient boosting and XGBoost in survival prediction

Gradient boosting frameworks configured for survival analysis have begun to reshape prognostic modeling in advanced cancer, particularly in settings where covariate effects are nonlinear and high dimensional. In stage IV non small cell lung cancer (NSCLC), XGBoost has been explicitly adapted to time to event data by employing a Cox-type loss function, thereby extending gradient boosting to handle censoring while retaining the interpretability of hazard-based formulations. Within this context, XGBoost is evaluated alongside Random Survival Forests (RSF), Cox proportional hazards (CoxPH) and single decision trees, using a consistent computational pipeline that emphasizes reproducible preprocessing, hyperparameter tuning and robust internal validation through repeated cross validation. Performance comparisons in this stage IV cohort demonstrate that ensemble-based methods provide measurable gains over traditional regression. Using repeated cross validation in a dataset with $n = 47$ observed events, the XGBoost survival model achieves a mean concordance index of 0.732 ± 0.041 and an Integrated Brier Score (IBS) of 0.124, indicating strong discrimination and relatively low time dependent prediction error. RSF attains slightly higher performance (C index 0.748 ± 0.036, IBS 0.118), while CoxPH reaches a C index of 0.718 ± 0.044 and IBS of 0.141, and a single decision tree performs worst (C index 0.658 ± 0.069, IBS 0.178). These metrics collectively highlight that XGBoost, when configured with a Cox survival objective, can better accommodate nonlinearities and higher order interactions among clinical, metastatic and treatment related variables than linear CoxPH, while maintaining competitive calibration as reflected by IBS. Hyperparameter optimization emerges as a central component of effective gradient boosting in this setting. Bayesian optimization via Optuna is used to tune key XGBoost parameters, including the number of estimators, learning rate, maximum tree depth and subsampling ratio. The optimal configuration identifies 735 estimators, a learning rate of 0.1402, maximum depth 3 and subsample ratio 0.7811. For RSF, tuning of $n_$estimators, minimum leaf size and feature sampling yields 642 trees, minimum leaf size 8 and $\log 2$ feature sampling. In contrast, CoxPH and decision tree models are deliberately retained with fixed parameter settings to preserve interpretability and ensure methodological comparability. Model reliability assessment using IBS confirms that the tuned XGBoost and RSF models provide the lowest prediction errors over the follow up period, underscoring the importance of systematic hyperparameter search when deploying gradient boosting for survival prediction. Analytical focus in this NSCLC study extends beyond discrimination metrics to encompass calibration and clinical utility. The framework integrates feature importance analyses to identify prognostically relevant variables and uses Kaplan–Meier risk stratification to examine separation between high risk and low risk groups defined by model-derived scores. Across CoxPH, decision tree, RSF and XGBoost models, pleural metastasis, liver metastasis and receipt of radiation therapy repeatedly emerge as critical predictors, indicating that organ specific metastatic burden and treatment covariates are consistently recognized as influential regardless of modeling approach. This concordance in feature importance, despite differences in algorithmic structure and flexibility, strengthens confidence that XGBoost-based survival models are capturing clinically meaningful prognostic signals rather than purely statistical artifacts. Limitations acknowledged in this work are directly relevant when considering extension of gradient boosting survival techniques to metastasis-related endpoints such as time to brain involvement or progression after metastasis. The dataset originates from a single institution, with modest sample size and limited events, constraining statistical power for assessing the independent contributions of individual metastatic locations and molecular biomarkers. External validation is absent, so generalizability of the XGBoost model to other cohorts and treatment eras remains uncertain. Potential confounders, including prior treatments, comorbidities, imaging characteristics and socioeconomic factors, are not captured, and comprehensive explainability analyses such as SHAP values or partial dependence plots fall outside the study scope. Nonetheless, the alignment of prognostic indicators across CoxPH and XGBoost, together with superior C index and IBS for gradient boosting, suggests that ensemble survival models can be integrated into clinically oriented pipelines, provided that future studies address sample size, external validation and interpretability requirements. Outside this stage IV NSCLC application, gradient boosting and related machine learning survival methods have been compared with CoxPH in early stage breast cancer and stage III colorectal cancer, though specific XGBoost configurations are not detailed. Those comparisons report only modest improvements in concordance indices and emphasize that many analyses focus predominantly on discrimination while underreporting calibration and net clinical benefit. Research therefore calls for expanded evaluation metrics and integrative frameworks that reconcile machine learning performance with clinical interpretability, an objective that the NSCLC study begins to address through its structured tuning, IBS assessment and feature importance analyses [Als+26].

7.2 Accelerated failure time models in metastasis research

Parametric survival frameworks such as accelerated failure time (AFT) models are conceptually well suited to describe nonlinear covariate effects on event times in metastatic disease, yet current evidence documents a striking paucity of explicit AFT applications for metastasis, recurrence or progression endpoints. Comparative work in oncology often lists AFT among candidate approaches but then implements Cox proportional hazards (CoxPH) and machine learning models as the primary tools, with little technical detail on AFT specifications or results. In early stage breast cancer, one study benchmarked AFT, Random Survival Forests and Gradient Boosting against CoxPH, reporting only marginal gains in concordance indices for machine learning methods and providing no description of the parametric distributions or time ratio estimates characteristic of AFT analysis. Similarly, stage III colorectal cancer investigations deployed deep learning Cox hybrids with concordance indices around 0.71 to 0.74, again focusing on discrimination rather than parametric modeling of survival time. Advanced non small cell lung cancer offers a more detailed methodological comparison, but the implemented models remain semi parametric or Cox based. In a stage IV cohort, survival prediction was explicitly framed as a comparison among CoxPH, decision trees, Random Survival Forests (RSF) and XGBoost configured with a Cox loss. The hazard function was modeled in standard Cox form, with the proportional hazards assumption checked using scaled Schoenfeld residuals and found to be adequately satisfied. RSF and XGBoost achieved higher concordance indices and lower Integrated Brier Scores than CoxPH and single decision trees, demonstrating the value of ensemble methods for handling nonlinearities and complex feature interactions, yet no AFT distributions were introduced and no parametric baseline hazards were estimated. Hyperparameter optimization via Optuna tuned RSF and XGBoost but left CoxPH with fixed penalization, underscoring that enhanced nonlinear modeling capacity was sought through tree based ensembles rather than through explicit AFT formulations [Als+26]. Clinical trial analyses in metastatic settings reinforce this pattern. For HER2 positive metastatic breast cancer treated with pyrotinib based regimens, progression free survival and overall survival were defined as times from treatment initiation to progression or death. Kaplan–Meier curves were used for visualization and log rank tests for subgroup comparisons, while Cox regression identified independent predictors of progression free survival, including number of metastatic sites and prior exposure to lapatinib. Again, parametric AFT models were not reported, and survival effects were summarized as hazard ratios rather than time ratios [Lin+20]. Prognostic nomograms for breast cancer brain metastases and lung cancer cohorts likewise rely on multivariable CoxPH as their statistical backbone. These tools quantify discrimination with Harrell's C index and assess calibration via bootstrap resampling, but they do not invoke Weibull, log normal or other AFT distributions, nor do they present covariate effects as multiplicative accelerations or decelerations of survival time [Hua+18; Yua+22].

Within this corpus, there is also no documentation of machine learning survival models that explicitly combine XGBoost with AFT distributions for metastasis related endpoints. XGBoost is described as a gradient boosting framework adapted to survival through a Cox type loss, with system optimizations such as shrinkage, column subsampling and regularization that make it suitable for high dimensional, sparse data. Applications to stage IV lung cancer adopt this Cox survival objective rather than an AFT loss, and outcomes are overall survival rather than time to metastasis, recurrence or progression. Other gradient boosting or deep learning efforts mentioned in oncology similarly employ Cox style hazards, leaving AFT based boosting for metastasis risk essentially unreported in the current evidence [Als+26]. Taken together, research shows that while AFT models are repeatedly acknowledged as potential alternatives to CoxPH, concrete implementations for organ specific metastasis, including brain involvement, are lacking in the available data. Survival modeling in metastatic oncology is presently dominated by semi parametric Cox frameworks and ensemble machine learning methods configured with Cox losses, with AFT and other parametric approaches remaining largely theoretical rather than practically deployed for metastasis, recurrence or progression endpoints [Als+26; Lin+20; Hua+18; Yua+22].

7.3 Integration of genomic and clinical data in survival models

Efforts to build survival models that are both prognostically accurate and biologically interpretable increasingly rely on integrating clinical covariates with genomic and metabolic information. In stage IV non small cell lung cancer, survival prediction has been systematically compared across Cox proportional hazards regression, decision trees, Random Survival Forests (RSF) and XGBoost configured with a Cox loss, using a unified preprocessing and validation pipeline. Clinical and metastatic variables such as pleural metastasis, liver metastasis and radiation therapy emerge as consistently important prognostic markers across all modeling frameworks, demonstrating that organ specific disease burden is a robust clinical axis of risk even when more flexible machine learning methods are used. These analyses are conducted in the context of high dimensional datasets where traditional Cox assumptions of proportional hazards and linear covariate effects may be violated, motivating incorporation of nonlinear modeling while retaining clinically interpretable covariates. Within this framework, ensemble machine learning methods are tuned to capture complex interactions among covariates, but the models still operate on clinically defined features like metastatic sites rather than directly on genomic profiles. RSF and XGBoost are optimized via Bayesian strategies implemented in Optuna, with RSF hyperparameters such as $n_$estimators, minimum leaf size and feature sampling, and XGBoost parameters including tree depth, learning rate, number of estimators and subsample ratio, carefully selected to balance predictive performance and overfitting. Reliability assessment using the Integrated Brier Score shows that RSF and XGBoost produce lower prediction error than Cox and decision trees, indicating that nonlinear modeling of clinical and metastatic features improves probabilistic survival prediction in advanced lung cancer. Nonetheless, Cox regression is retained with fixed penalization to preserve transparency, and decision trees remain in the model set to provide easily interpretable rules, underscoring an explicit strategy to combine performance oriented machine learning with interpretable regression [Als+26]. Parallel work in breast cancer demonstrates how molecular markers are linked to survival analyses, even though full machine learning integration is not yet described. In a pyrotinib based trial for HER2 positive metastatic breast cancer, progression free survival is modeled using Kaplan–Meier curves and Cox multivariate regression, with the number of metastatic sites and prior exposure to lapatinib identified as independent predictors of progression free survival. Brain metastasis status is included as a clinical variable, and overall median progression free survival for patients with brain metastases is reported, but the statistical modeling remains purely Cox based without ensemble or gradient boosting components. This example illustrates how clinical risk scores and Cox models currently dominate survival analyses even when detailed treatment history and metastatic burden are incorporated, and highlights a gap in explicit integration of genomic determinants of HER2 inhibitor resistance, such as DUSP6 mediated neratinib resistance, into time to event models [Lin+20; Mom+23]. Beyond HER2 targeted therapy, autophagy and PI3K/AKT pathway activation are recognized as key survival mechanisms in breast cancer, and agents such as everolimus plus hydroxychloroquine, or resveratrol, are discussed as modulators of autophagic flux and metabolic state. These pathways are genetically and biochemically defined, yet they have not been coupled in the available evidence to machine learning survival models that could quantify how autophagy related biomarkers modify time to progression or brain metastasis. Instead, autophagy targeting is motivated by mechanistic data and evaluated in clinical trials using conventional survival endpoints analyzed with Cox regression, reinforcing the current separation between molecular pathway elucidation and advanced survival modeling [MH21]. Research on stage IV lung cancer explicitly calls for frameworks that reconcile methodological rigor with clinical relevance by integrating feature importance analyses, expanded evaluation metrics, and reproducible tuning. However, genomic data are not incorporated in the reported models, and XGBoost is used only with a Cox survival objective rather than accelerated failure time formulations. There is no documented application combining XGBoost with AFT distributions or directly modeling time to metastasis, recurrence, or progression using integrated genomic and clinical covariates. Instead, the existing work shows that ensemble Cox based boosting and RSF can effectively capture nonlinear survival patterns driven by clinical and metastatic variables, and advocates future multi center, molecularly annotated datasets to extend these approaches toward genuine genomic–clinical integration in survival prediction [Als+26].

7.4 Evaluation metrics and validation strategies for time-to-event models

Quantitative assessment of time-to-event models in metastatic oncology relies on a limited but coherent set of discrimination, calibration and validation tools. In advanced non small cell lung cancer, discrimination is most commonly summarized by the concordance index (C index), applied uniformly across Cox proportional hazards (CoxPH), Random Survival Forests (RSF), XGBoost with a Cox loss and single decision trees. In a stage IV cohort with $n = 47$ events, RSF achieved a mean C index of 0.748 ± 0.036, XGBoost 0.732 ± 0.041, CoxPH 0.718 ± 0.044 and the decision tree 0.658 ± 0.069, providing a clear ranking of models in terms of discriminative ability. Similar use of Harrell's C index appears in nomogram based work for breast cancer brain metastases and for lung cancer cohorts, where it measures how well the model separates patients who die from those who survive over specified horizons [Als+26; Hua+18; Yua+22]. Calibration and overall prediction error are evaluated through the Integrated Brier Score (IBS) in the same lung cancer study. IBS jointly reflects discrimination and calibration over the follow up period in the presence of censoring. RSF yields the lowest IBS at 0.118, closely followed by XGBoost at 0.124, while CoxPH and the decision tree reach 0.141 and 0.178, respectively. These values underpin the conclusion that ensemble machine learning models provide more accurate probabilistic survival predictions than traditional and single tree approaches in that advanced stage cohort. Model reliability assessment explicitly relies on IBS rather than fixed time calibration plots alone, thereby avoiding sensitivity to censoring at particular time points [Als+26]. Graphical calibration is nevertheless used in nomogram development for breast cancer brain metastasis and for lung cancer. In the breast cancer brain metastasis nomogram, calibration curves compare predicted probabilities from the Cox based model with observed brain metastasis overall survival (BMOS) across risk strata, after internal validation by 1000 fold bootstrap resampling. For the lung cancer nomogram, calibration plots similarly assess agreement between predicted and observed overall survival, and the C index is again used to compare models. These procedures provide a complementary view to IBS, focusing on the extent to which predicted survival probabilities match empirical outcomes across the risk spectrum [Hua+18; Yua+22]. Internal validation strategies differ between studies but share an emphasis on resampling. The lung cancer machine learning analysis employs repeated cross validation to reduce variability associated with single train test splits, an approach chosen because of the limited sample size and event count. Performance metrics in this setting, including C index and IBS, are averaged across repetitions to obtain stable estimates. In contrast, the breast cancer brain metastasis nomogram relies on bootstrap resampling to internally validate the Cox model, again focusing on discrimination and calibration. Both approaches aim to mitigate overfitting in modest datasets, although neither includes external validation [Als+26; Hua+18]. Hyperparameter optimization is treated as an integral part of validation for ensemble models. In the lung cancer study, Bayesian optimization via Optuna tunes RSF parameters such as $n_$estimators, minimum leaf size and feature sampling, and XGBoost parameters including learning rate, depth, estimator number and subsample ratio. The selected configurations are then evaluated under repeated cross validation, with IBS and C index serving as primary metrics. CoxPH and decision tree models are kept with fixed parameter settings to preserve interpretability and to allow fair comparison with tuned ensembles. This strategy yields what the authors describe as stable and unbiased performance estimation for RSF and XGBoost, while maintaining transparency in model configuration. Proportional hazards diagnostics enter the evaluation toolbox for Cox based methods. In the same lung cancer cohort, scaled Schoenfeld residuals are used to assess the proportional hazards assumption, and no significant violations are reported, supporting the use of

CoxPH as a benchmark. Nonetheless, the analysis notes that in complex high dimensional datasets, proportional hazards and linear covariate effects are often extensively violated, which motivates the inclusion of RSF and XGBoost and the reliance on C index and IBS to compare models under more flexible assumptions [Als+26]. Across these studies, evaluation of time-to-event models concentrates on discrimination via C index, calibration via IBS or calibration plots, and internal validation through cross validation or bootstrap methods. External validation, decision curve analysis and explainability tools such as SHAP values or partial dependence plots are explicitly identified as missing and proposed as priorities for future work, particularly when ensemble machine learning models are applied to survival prediction in advanced metastatic cohorts [Als+26; Hua+18; Yua+22].

8 Linking mechanistic biology with predictive modeling

8.1 From molecular mechanisms to model covariates

Translating mechanistic insights into quantitative covariates for time-to-event models requires first identifying which biological processes are robustly supported by data and then mapping them to measurable features. In stage IV non-small-cell lung cancer, for example, survival models that include metastatic site indicators such as pleural and liver involvement achieve strong prognostic discrimination, and these variables consistently emerge as important across Cox, decision tree, Random Survival Forest (RSF) and XGBoost models. This convergence indicates that organ-specific metastatic burden is a stable candidate covariate regardless of the modeling framework and can be treated as a clinically grounded surrogate for complex organ-level biology in survival prediction [Als+26]. Mechanistic work on breast cancer brain metastasis provides a complementary template for covariate design. Metabolic reprogramming in brain-tropic breast cancer cells involves upregulation of glycolysis, tricarboxylic acid cycle, oxidative phosphorylation and glutathione pathways, together enabling oxidative glucose metabolism and enhanced redox control in the energy-constrained brain parenchyma. In parallel, high expression of fatty acid binding protein 7 supports adaptation of HER2 cells by sustaining glycolysis and lipid droplet storage, while brain metastatic cells exploit gluconeogenesis, branched-chain amino acid oxidation and acetate metabolism. These findings justify constructing covariates that capture expression or activity of oxidative phosphorylation, lipid handling proteins such as FABP7, and amino acid metabolic pathways when modeling survival or progression in cohorts enriched for brain metastasis [WL21]. Autophagy-related mechanisms constitute another mechanistic axis that can be directly reflected in covariates. Overexpression of the glucose-regulated chaperone GRP94 relieves metabolic stress under low-glucose conditions by coordinating the unfolded protein response and pro-survival autophagy in breast cancer brain metastases, and KISS1-mediated survival autophagy has been proposed as a diagnostic marker and therapeutic target in this setting. In addition, regulatory nodes such as BECN1, FIP200, p53 and YAP are implicated in breast tumour initiation, progression and metastasis via autophagy modulation. These observations support the inclusion of variables capturing expression or mutational status of GRP94, KISS1 and core autophagy regulators in models aimed at predicting outcomes for patients with brain involvement, particularly when therapies targeting autophagy are being evaluated [MH21]. Time-to-event modeling in lung cancer offers a methodological scaffold for embedding such covariates. RSF and XGBoost configured with a Cox survival objective deliver higher concordance indices and lower Integrated Brier Scores than Cox proportional hazards and single decision trees in a stage IV lung cancer cohort, despite the modest sample size. Hyperparameter tuning via Bayesian optimization (Optuna) refines tree number, depth, learning rate and subsampling ratios, yielding RSF and XGBoost configurations that provide stable, unbiased prediction error estimates. Within this tuned framework, pleural and liver metastases and radiation therapy remain dominant features, underscoring that biologically interpretable covariates preserve their relevance even in complex ensemble models [Als+26]. Experimental metastasis studies further illustrate how experimental endpoints can be shaped into time-to-event outcomes. In a prophylactic cranial irradiation model, the incidence of brain metastasis at an eight-week endpoint varies sharply with the timing of cranial radiation, with early irradiation substantially lowering the proportion of mice with brain lesions compared to non-irradiated or differently timed regimens. Although these data are reported as incidence proportions rather than survival curves, they imply that timing of cranial irradiation could be modeled as a categorical covariate in future time-to-metastasis analyses of similar designs [Smi+21]. Clinical trials in metastatic breast cancer show how treatment, metastatic pattern and survival are routinely connected via Cox regression and Kaplan–Meier analyses. In HER2-positive metastatic disease treated with pyrotinib-based regimens, progression-free survival and overall survival are estimated non-parametrically, while Cox regression identifies independent predictors such as the number of metastatic sites and prior lapatinib exposure. These predictors are readily interpretable mechanistic proxies: metastatic site number reflects systemic tumour burden, and prior HER2-targeted therapy captures pre-existing pathway inhibition and resistance. Incorporating such variables alongside mechanistically informed biomarkers, for example autophagy or metabolic markers, would extend current Cox-based frameworks into more biologically grounded survival models [Lin+20]. Nomogram-based risk tools demonstrate a parallel pathway from covariates to clinical decision support. In lung cancer, age, marital status, sex, race, grade, T stage and primary tumour site are combined into a Cox-based nomogram that predicts overall survival, with internal validation via concordance index and calibration plots. A similar strategy applied to breast cancer brain metastasis uses Cox-derived hazard ratios to build a nomogram for brain metastasis overall survival, again validated by Harrell's C index and bootstrap calibration. These examples confirm that once mechanistically motivated covariates are identified and quantified, they can be embedded into both traditional nomograms and modern ensemble survival models to bridge detailed biology with actionable prognostic tools [Yua+22; Hua+18; Als+26].

8.2 Modeling nonlinear effects of mutations and pathways on survival

Nonlinear survival effects arising from molecular pathways are most clearly documented where mechanistic data intersect with flexible prognostic modeling. In advanced non small cell lung cancer, ensemble survival methods such as Random Survival Forests and XGBoost configured with a Cox loss outperform traditional Cox proportional hazards and single decision trees when predicting time to death in a stage IV cohort. Using repeated cross validation with $n = 47$ events, Random Survival Forest achieves a concordance index of 0.748 ± 0.036 and Integrated Brier Score (IBS) of 0.118, while XGBoost attains 0.732 ± 0.041 and 0.124, respectively, compared with 0.718 ± 0.044 and 0.141 for Cox and 0.658 ± 0.069 and 0.178 for a single decision tree. These differences indicate that tree based ensembles capture nonlinear and interaction effects among clinical, metastatic and treatment covariates that are not well represented by linear hazard structures. Model configuration in this lung cancer study directly addresses nonlinearity. Random Survival Forests are tuned via Bayesian optimization of tree number, minimum leaf size and feature sampling, yielding an optimal setting of 642 trees, minimum leaf size 8 and $\log 2$ feature sampling. XGBoost with a Cox survival objective is similarly optimized for 735 estimators, learning rate 0.1402, maximum depth 3 and subsample ratio 0.7811. In contrast, Cox models are kept with fixed penalization to preserve interpretability. The lower IBS values for the tuned ensemble models demonstrate that explicitly modeling nonlinear feature effects improves probabilistic survival prediction in the presence of complex covariate structures, even though the underlying hazard formulation remains Cox type rather than accelerated failure time. Importantly, pleural metastasis, liver metastasis and radiation therapy appear as key prognostic indicators across all four models, showing that nonlinearity is captured around clinically meaningful variables rather than around arbitrary features [Als+26]. Applications directly targeting metastasis or progression endpoints rely primarily on Cox based approaches, but they still embody nonlinear behavior when combined with stepwise variable selection and nomogram construction. In breast cancer patients with brain metastases, overall survival after brain involvement is modeled using Cox regression, with significant variables from univariate analysis entered into multivariable models and independent factors selected via backward stepwise procedures. These coefficients form the basis of a nomogram for brain metastasis overall survival, validated internally using Harrell's C index and bootstrap calibration curves. Although the statistical backbone is proportional hazards, the combination of categorical and continuous covariates, and their interactions in the nomogram, creates a piecewise nonlinear mapping from biomarker and clinical status to survival probability [Hua+18]. A similar strategy in lung cancer combines age, marital status, sex, race, tumour grade, T stage and primary site into a Cox based nomogram. Here, higher age, advanced grade and T stage act as risk factors, while marital status, female sex and white race are protective. The resulting tool assigns weighted scores to each covariate and maps total score to predicted survival. Internal validation again uses the C index and calibration plots. Even without explicit nonlinear survival distributions, this approach implements a structured, multi parameter transformation from covariate space to survival, effectively modeling complex effects of host and tumour factors on outcome [Yua+22]. Machine learning based survival frameworks broaden the capacity to encode nonlinear contributions of mutations and pathways, but explicit integration of genomic features remains limited in the available evidence. Research on ferroptosis related gene signatures in breast cancer establishes a multigene risk score via univariate and multivariate Cox regression on ferroptosis regulators, defining high and low risk groups according to a linear combination of standardized expression values. Single sample gene set enrichment analysis is used to link ferroptosis genes to immune cell infiltration and pathway activity, and a ferroptosis score derived from principal component analysis separates patients into groups with distinct tumour microenvironment characteristics. External validation on datasets containing primary and brain metastatic breast cancer tissues demonstrates differential expression of ferroptosis regulators in brain metastases, and Kaplan–Meier analysis in The Cancer Genome Atlas confirms prognostic effects of ferroptosis patterns. These studies quantify survival influences of pathway level gene programs, although they deploy Cox based rather than accelerated failure time models and do not yet embed these signatures within ensemble methods like XGBoost [Zhu+22]. Across these contexts, accelerated failure time and other fully parametric nonlinear survival models are acknowledged conceptually but are not implemented in detail for metastasis, recurrence or progression endpoints. Comparative modeling in oncology notes AFT as an alternative to Cox, yet practical applications for organ specific metastasis, including brain involvement, are not described. Likewise, there is no documented use of XGBoost combined with AFT distributions for time to metastasis or progression; gradient boosting in stage IV lung cancer is realized solely through a Cox loss. Thus, current evidence demonstrates that nonlinear effects of mutations, pathways and metastatic sites on survival are being captured predominantly through Cox based ensemble methods, Cox derived nomograms and multigene risk scores, rather than through explicit parametric AFT formulations [Als+26; Hua+18; Yua+22; Zhu+22].

8.3 Bridging experimental and clinical data for metastasis modeling

Experimental work on organ-specific metabolism and survival can be linked to clinical outcome modeling by treating mechanistically defined pathways as candidate prognostic features and then quantifying their impact with time-to-event methods. In advanced lung cancer, survival prediction has been benchmarked across Cox proportional hazards, decision trees, Random Survival Forests (RSF) and XGBoost configured with a Cox loss, using a unified computational pipeline. Clinical and metastatic covariates such as pleural metastasis, liver metastasis and radiation therapy repeatedly emerge as important prognostic markers across all four models, underscoring that organ-specific disease burden is a robust axis of risk even when complex machine learning methods are applied [Als+26]. These variables encapsulate aspects of systemic tumour biology and could, in principle, be complemented by mechanistically grounded biomarkers, for example metabolic or autophagy-related measures, in future model extensions. Mechanistic studies on breast cancer brain metastasis provide clear examples of such candidate biomarkers. Brain-tropic breast cancer cells upregulate enzymes in glycolysis, the tricarboxylic acid cycle, oxidative phosphorylation, pentose phosphate pathway and glutathione system, allowing simultaneous oxidative glucose metabolism and enhanced antioxidant defence in the energy-constrained brain parenchyma. High expression of fatty acid binding protein 7 promotes adaptation of HER2 cells by supporting a glycolytic phenotype and lipid droplet storage, while brain metastatic cells can use gluconeogenesis, branched-chain amino acid oxidation and acetate metabolism to sustain growth. These findings delineate metabolic traits that could be operationalized as covariates in survival or progression models for patients with brain metastases, provided suitable clinical assays are available [WL21]. Autophagy-related mechanisms, such as overexpression of glucose-regulated protein GRP94 and KISS1-mediated survival autophagy, likewise define stress-response programmes that are directly relevant to brain metastatic fitness and thus natural candidates for inclusion in prognostic modeling [MH21]. Bringing these biological determinants into survival frameworks requires attention to model choice and evaluation. In the stage IV non-small-cell lung cancer cohort, RSF and XGBoost (survival: Cox) outperform Cox proportional hazards and single decision trees, achieving higher concordance indices and lower Integrated Brier Scores under repeated cross-validation. RSF reaches a mean C index of 0.748 ± 0.036 with IBS $= 0.118$, XGBoost attains 0.732 ± 0.041 and 0.124, whereas Cox yields 0.718 ± 0.044 and 0.141 and the decision tree 0.658 ± 0.069 and 0.178. Hyperparameter optimization via Bayesian strategies (Optuna) tunes RSF parameters such as n_estimators, minimum leaf size and feature sampling, and XGBoost parameters including learning rate, depth, estimator number and subsample ratio, while Cox and decision trees are kept fixed to preserve interpretability. Reliability assessment with IBS confirms that ensemble methods provide more accurate probabilistic survival predictions, suggesting that similarly tuned RSF and XGBoost models could be used to evaluate the prognostic impact of brain-metastasis-specific metabolic and autophagy markers once integrated into clinical datasets [Als+26]. Current work on ferroptosis-related genes in breast cancer illustrates a partial bridge between molecular signatures and survival analysis. A multigene ferroptosis risk score, derived from 14 ferroptosis-related genes, stratifies patients into high- and low-risk groups with significantly different overall and relapse-free survival; higher risk scores associate with increased death probability and shorter survival times. Expression of individual ferroptosis regulators such as ACACA, DPP4, GSS and HMOX1 shows subtype-specific prognostic effects, and several genes display strong associations with tumour grade, estrogen receptor and progesterone receptor status. These analyses use Cox-based hazard modeling and Kaplan–Meier curves to connect gene expression patterns to clinical outcomes, providing a template for how metabolism-related pathways could be quantitatively linked to metastasis risk, even though brain-specific endpoints and machine learning survival models are not yet applied in this context [Zhu+22]. Clinical trials in metastatic breast cancer demonstrate that treatment variables and metastatic burden are already embedded in Cox regression models. In HER2-positive metastatic disease treated with pyrotinib-based regimens, the number of metastatic sites and prior exposure to lapatinib are independent predictors of progression-free survival in Cox multivariate analysis, with hazard ratios of 1.778 and 2.313, respectively. Brain metastasis status is included descriptively, with median progression-free survival of 6.7 months for patients with baseline brain involvement, but is not modeled as an independent covariate. These results show how discrete clinical features reflecting systemic disease burden and prior pathway inhibition are mapped to survival outcomes, and highlight the opportunity to supplement them with mechanistic markers, such as PHGDH or FABP7 expression, in future modeling efforts [Lin+20]. Across these studies, survival modeling for metastasis, recurrence or progression relies on Cox-based and tree-ensemble frameworks, with RSF and XGBoost (using a Cox loss) providing improved handling of nonlinear clinical effects. Explicit accelerated failure time models or XGBoost combined with AFT distributions are not documented, and genomic or metabolic biomarkers are only beginning to be connected to survival through multigene risk scores and conventional regression. Bridging experimental and clinical data for metastasis modeling will therefore depend on assembling molecularly annotated cohorts and extending the demonstrated RSF and XGBoost pipelines to incorporate organ-specific metabolic and stress-response covariates alongside established clinical predictors [Als+26; WL21; MH21; Zhu+22].

9 Methodological considerations and limitations

9.1 Data sources for metastasis and survival studies

Data sources underpinning metastasis and survival studies in this thesis span institutional cohorts, population registries and multi platform genomic repositories, each imposing distinct constraints on inference. In advanced non small cell lung cancer, a single institution dataset was assembled to compare Cox proportional hazards, decision tree, Random Survival Forest and XGBoost survival models. This cohort comprised stage IV cases with $n = 47$ observed events, and incorporated demographic, clinical and metastatic variables such as pleural and liver involvement and receipt of radiation therapy. Survival times and censoring were derived from routine clinical follow up, with imaging and treatment records providing covariates. The modest sample size and single center origin limited statistical power for estimating independent effects of individual metastatic locations and molecular biomarkers, and the absence of external validation restricted generalizability, highlighting typical weaknesses of institution based survival datasets in metastasis research [Als+26]. Another clinically grounded source focuses on resected brain metastasis cavities treated with adjuvant Gamma Knife radiosurgery. Here, data acquisition was cavity centered: postoperative cavity volumes were segmented on serial contrast enhanced MRI, log transformed because of right skew, and analyzed using linear mixed effects models with an AR(1) covariance structure to capture within patient serial correlation. Local recurrence was defined radiographically as progressive or new nodular enhancement within or adjacent to the treated cavity, with contrast clearance analysis used to adjudicate equivocal cases. Local recurrence free survival was measured from radiosurgery to first imaging evidence of recurrence, with cavities censored at last intracranial imaging or death without recurrence. Kaplan–Meier methods yielded a median local recurrence free survival of 49.8 months and declining local control estimates at successive time points, but decreasing radiographic surveillance over time reduced the at risk set and widened uncertainty at later follow ups. This design illustrates how detailed imaging based local endpoints, coupled to institutional follow up patterns, shape survival analyses in brain metastasis cohorts [Gou+26]. Population level registries such as the Surveillance, Epidemiology, and End Results database provide a contrasting data source. In one study, SEER records were used to build a nomogram for patients with lung cancer and simple brain metastases. Available variables included age, race, gender, histologic grade, TNM staging and limited treatment information, but genomic status, protein expression and family history were absent. Radiotherapy and chemotherapy fields were incomplete and heavily influenced by patient treatment preferences, making them unsuitable for robust model construction. The retrospective nature of SEER based analyses and the lack of local validation data for other regions further constrained applicability, yet the large sample size allowed identification of prognostic factors and construction of a broadly calibrated nomogram despite these limitations [Yua+22]. Hybrid clinical trial datasets contribute additional survival information, especially for metastatic breast cancer. In a phase I/II study of the Esperanza extract in metastatic gastrointestinal tumours and acute leukemias, outcomes included progression free survival and overall survival alongside quality of life measures. Progression free survival was defined from randomization to disease progression, local or distant recurrence, second primary malignancy or death, with patients censored if they did not meet the evaluation time point and followed until end of treatment for survival curve preparation. Sample sizes were determined by predefined effect sizes and error rates, and survival comparisons were analyzed using log rank tests. Although not focused on brain metastasis, this protocol exemplifies how experimental therapeutics trials generate structured time to event data for metastatic populations [Bal+23]. Genomic and transcriptomic resources expand data sources beyond purely clinical registries. For NSCLC brain metastases, transcriptome, clinical survival and characteristic information were collected from The Cancer Genome Atlas (LUAD and LUSC) and multiple Gene Expression Omnibus microarray datasets, with an additional NSCLC dataset containing brain metastasis features. Quality control steps removed samples lacking follow up, converted gene names to uniform identifiers, merged datasets and removed batch effects using dedicated R functions. Univariate Cox regression, followed by multivariate Cox and LASSO analyses, identified an 11 gene prognostic signature for NSCLC brain metastases and supported construction of a RiskScore model validated against immune microenvironment, drug sensitivity, tumour mutation and decision tree analyses. This integrated multi cohort design illustrates how publicly available omics and clinical survival data can be curated for organ specific metastasis modeling, though it depends on consistent annotation across platforms and careful batch correction [Wan+23]. Finally, experimental metastasis models generate survival data at the animal level. In preclinical studies of acetate metabolism and acetyl CoA synthetase 2 inhibition in breast cancer brain metastasis, mice injected with luciferase tagged brain tropic 4T1 cells were treated with ACSS2 inhibitors or vehicle, and bioluminescence imaging tracked intracranial tumour burden over time. Kaplan–Meier survival analysis quantified differences between treatment and control groups, with significant prolongation of survival in inhibitor treated mice. These animal based datasets provide mechanistic insight into metabolic pathways relevant to brain metastasis and generate structured survival endpoints, but their small group sizes and species specific physiology limit direct translation to human risk modeling [Esq+24].

9.2 Biases and confounding in prognostic modeling

Bias and confounding arise from both data sources and modeling choices in prognostic analyses of metastasis and survival. Single institution cohorts in stage IV non small cell lung cancer exemplify several structural limitations. The dataset used to compare Cox proportional hazards, decision tree, Random Survival Forest and XGBoost contained only $n = 47$ observed events, constraining statistical power for independent effects of metastatic locations and molecular biomarkers. Absence of an external validation cohort restricts generalizability, and important potential confounders such as prior treatment history, comorbidity burden, detailed radiologic characteristics and socioeconomic variables were not recorded, meaning that estimated effects of pleural metastasis, liver metastasis or radiation therapy may be partially confounded by unmeasured factors. Temporal bias is also acknowledged because treatment modalities and supportive care evolved over the study period, so survival differences can reflect changing practice patterns rather than covariates alone. Although scaled Schoenfeld residuals showed no major violation of the proportional hazards assumption, minor breaches cannot be excluded given the modest sample size, adding another layer of uncertainty to Cox based inference. Use of ensemble machine learning methods introduces a different class of bias risks. Random Survival Forest and

XGBoost configured with a Cox loss achieved higher concordance indices and lower Integrated Brier Scores than Cox and single decision trees under repeated cross validation, supporting their predictive superiority. However, these models are described as "black box" strategies, and comprehensive explainability evaluations such as SHAP values or partial dependence plots were explicitly outside the study scope. This opacity can produce interpretability bias, where clinicians may undertrust or misinterpret risk estimates, and it complicates identification of spurious associations arising from complex feature interactions. Hyperparameter optimization via Bayesian methods (Optuna) reduces overfitting risk, yet tuning on a small, single center dataset can still lock in idiosyncratic patterns that do not generalize, particularly when external validation is missing. Future multi center datasets with richer clinical and molecular annotations are therefore deemed essential to mitigate these biases and confounding structures [Als+26]. Registry based nomogram studies for lung cancer and breast cancer brain metastases face complementary challenges. Analyses using Surveillance, Epidemiology, and End Results data for lung cancer rely on variables such as age, marital status, sex, race, grade, T stage and primary site, but lack genomic profiles, protein expression and detailed treatment information. Radiotherapy and chemotherapy entries are incomplete and influenced by patient preference, so treatment effects in the nomogram are susceptible to selection bias and confounding by indication. Internal validation with Harrell's C index and calibration plots cannot compensate for the absence of external validation or adjustment for unmeasured biological factors. Similarly, the breast cancer brain metastasis nomogram is constructed on retrospective institutional data, with multiple exclusion criteria and reliance on imaging or pathology for brain metastasis diagnosis. Although bootstrap resampling is used to minimize optimism, selection of patients from a single hospital and omission of granular molecular data make the model vulnerable to institutional practice bias and residual confounding in estimated brain metastasis overall survival [Yua+22; Hua+18]. Trial based survival analyses incorporate their own sources of bias. In HER2 positive metastatic breast cancer treated with pyrotinib, progression free survival is modeled via log rank and Cox multivariate analysis, identifying number of metastatic sites and prior lapatinib exposure as independent predictors. However, prior exposure reflects both biological resistance mechanisms and treatment history decisions, and without randomization across these strata, hazard ratios for prior lapatinib are inherently confounded by clinical judgment and disease course. Brain metastasis status is described, with median progression free survival of 6.7 months in patients with baseline brain involvement, but not adjusted in multivariable models, raising the possibility that its influence is entangled with other metastatic burden and therapy variables. Moreover, the absence of accelerated failure time or other parametric approaches leaves potential nonlinear time scaling effects unexplored, and reliance on a single Cox framework may mask time dependent treatment effects [Lin+20]. Finally, studies that compare Cox, AFT, RSF, Gradient Boosting or deep Cox hybrids in other cancers often emphasize discrimination (C index, AUC) while underreporting calibration, Brier score or decision curve analysis. This evaluation imbalance can bias perceptions of model utility, as modest gains in discrimination may be overstated without parallel assessment of calibration and clinical net benefit. In aggregate, these methodological patterns show that prognostic modeling for metastasis and survival is frequently shaped by sample selection, missing covariates, evolving treatments, limited validation and incomplete evaluation metrics, all of which contribute to bias and confounding that must be explicitly acknowledged and addressed [Als+26; Hua+18; Yua+22; Lin+20].

9.3 Reproducibility and generalizability of predictive models

Evidence from advanced non small cell lung cancer illustrates both strengths and limits of reproducibility in survival prediction pipelines. In a stage IV cohort used to compare Cox proportional hazards, decision tree, Random Survival Forest and XGBoost survival models, internal validation relied on repeated cross validation, a strategy chosen to reduce variability from single train–test splits in the presence of only $n = 47$ observed events. This framework produced stable estimates of concordance indices and Integrated Brier Scores, with RSF and XGBoost consistently outperforming Cox and single trees, suggesting that the tuning and evaluation procedures are reproducible within the dataset. However, the same analysis explicitly acknowledges that generalizability is constrained by the single-institution origin of the data and the absence of external validation, limiting confidence that the superior performance of ensemble models will hold across other populations, treatment eras or metastatic patterns. Hyperparameter optimization in this cohort further shapes reproducibility. Bayesian optimization via Optuna was applied to RSF and XGBoost, exploring predefined ranges for parameters such as n_estimators, minimum leaf size, feature sampling strategy, learning rate, depth and subsample ratio. Selected configurations, including 642 trees and $\log 2$ feature sampling for RSF and 735 estimators with learning rate 0.1402 for XGBoost, were then evaluated under repeated cross validation. Authors report that this tuning framework yielded "stable and unbiased" performance estimation, indicating that given the same data and search ranges, similar hyperparameter sets and predictive metrics are likely to be recovered, which supports methodological reproducibility. Yet the fact that Cox and decision tree models were kept with fixed parameter settings, partly for interpretability, introduces asymmetry between tuned and untuned models that may complicate generalization of relative performance rankings to other datasets where tuning choices differ. Model reliability assessment via Integrated Brier Score contributes another dimension. RSF achieved the lowest IBS (0.118), followed by XGBoost (0.124), whereas Cox and the decision tree showed higher prediction errors (0.141 and 0.178). These results argue that ensemble methods produce more accurate probabilistic survival predictions over the follow up period in this lung cancer cohort, but they are derived entirely from internal resampling. Without external test sets, it remains unknown whether IBS advantages persist when models are applied to new institutions, different distributions of metastatic sites such as brain versus pleural disease, or distinct treatment regimes. Limitations discussed in the same work speak directly to generalizability. The modest sample size and limited number of events restrict statistical power, particularly for estimating independent impacts of metastatic locations and molecular biomarkers. Unmeasured confounders, including prior therapy, comorbidities, detailed radiologic characteristics and socioeconomic status, were not available, so survival associations with pleural metastasis, liver metastasis or radiation therapy may partly reflect unrecorded factors. Temporal bias is introduced by evolving treatment modalities and supportive care during the study period, potentially embedding era specific effects into learned models. Finally, RSF and XGBoost are described as "opaque" black box strategies, with comprehensive explainability analyses such as SHAP values and partial dependence plots left for future work, which complicates assessment of whether learned nonlinear relationships would be trustworthy when transferred to other settings [Als+26]. Nomogram based studies in lung and breast cancer show analogous tensions between internal consistency and external transportability. A lung cancer nomogram built from SEER registry data uses Cox regression to combine age, marital status, sex, race, grade, T stage and primary site into an overall survival prediction tool, with discrimination quantified by C index and calibration assessed graphically. However, missing genomic and treatment data and patient preference driven radiotherapy and chemotherapy choices limit control of confounding and hinder generalization beyond the registry context [Yua+22]. Similarly, a prognostic nomogram for breast cancer brain metastases is internally validated by bootstrap resampling and Harrell's C index, but it is derived from a single hospital's retrospective cohort, again without external validation or detailed molecular annotation, compromising broader applicability despite good internal calibration [Hua+18]. Taken together, current survival modeling efforts in metastatic oncology demonstrate reproducible internal pipelines with repeated cross validation, bootstrap resampling and clearly specified hyperparameter optimization, but generalizability is curtailed by single center data, limited event numbers, missing covariates and lack of external validation. Ensemble machine learning methods such as RSF and XGBoost offer improved discrimination and calibration in advanced lung cancer, yet their transport to other institutions, cancer types or organ specific metastasis endpoints will require multi center, molecularly annotated datasets and explicit external performance assessment [Als+26; Hua+18; Yua+22].

10 Conclusion

This work synthesises mechanistic and methodological threads to argue that organ specific metastasis, and brain metastasis in particular, is best approached through integrated genomic, metabolic and time to event modeling. Experimental studies identify recurrent metabolic programmes in braintropic breast cancer , including PHGDH dependent de novo serine synthesis, enhanced glutamine and branched chain amino acid oxidation, FABP7 linked lipid handling, acetate and gluconeogenic routes, and GRP94 or KISS1 mediated autophagy , while genomic alterations such as PTEN loss, RARRES2 suppression and PI3K pathway activation align with SREBP1 driven lipogenesis and altered lipid profiles. At the same time, clinical modeling in advanced NSCLC demonstrates that ensemble survival methods (Random Survival Forests, XGBoost with a Cox loss) tuned by Bayesian search (Optuna) yield higher concordance indices and lower Integrated Brier Scores than CoxPH in modest cohorts ($n = 47$ events), with pleural and liver metastases and radiation therapy repeatedly emerging as influential predictors. These complementary strands imply that mechanistically informed biomarkers (for example PHGDH, FABP7, GRP94, ferroptosis regulators, ADAM22–LGI1 axis) can be translated into covariates for ensemble survival frameworks, while acknowledging that Cox based nomograms retain interpretability for clinical decision support.

To translate these insights into robust prognostic tools requires coordinated data generation and careful modeling practice. Priority actions include assembling multi center, molecularly annotated cohorts with standardized assays for metabolic and autophagy markers, rigorous external validation of tuned RSF and XGBoost pipelines, explicit calibration reporting (IBS and calibration plots) alongside discrimination (C index), and use of explainability methods such as SHAP or partial dependence plots to probe learned nonlinear interactions. Where biologic rationale suggests time scaling effects, parametric alternatives such as AFT models should be explored in parallel to Cox based losses and boosted implementations, with attention to censoring and sample size constraints. Integrative efforts that couple experimental mechanistic endpoints with reproducible machine learning survival pipelines offer the most direct route to quantify how tumour genomic and metabolic programmes shape organ specific colonization and patient trajectories, and to identify context dependent therapeutic vulnerabilities that are measurable and actionable in clinical cohorts.

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